

A constraint on quantification over parts

Abstract

This paper investigates why some predicates are compatible with quantification over parts of singular individuals, while others are not. For example, *Most of the trail is narrow* is acceptable, but *#Most of the trail is short* is not. I propose a constraint on quantification over parts, which derives the contrast between quantification-compatible and quantification-resistant predicates from an independently motivated pragmatic principle banning contextual equivalence among scalar alternatives. This analysis explains when and why quantification over parts of singular individuals gives rise to infelicity, and it naturally extends to the phenomenon of quantification compatibility in collective predication, known as the *gather/numerous* distinction.

Keywords: Subatomic quantification, Partitives, Mereology, Collective predication, Contextual equivalence

1 Introduction

Not all predicates are compatible with quantification over parts of a singular individual (Löbner 2000; Corblin 2008).¹ Some predicates readily combine with partitive quantifiers such as *part*, *most*, or *all* (1), whereas other predicates cannot co-occur with these quantifiers (2).

- (1) a. Part of the trail is very narrow.
b. Most of the rug is ruined.
- (2) a. #Part of the trail is very short.
b. #Most of the rug really tied the room together.

I refer to predicates that permit quantification over parts as QUANTIFICATION-COMPATIBLE, and those that do not as QUANTIFICATION-RESISTANT.

Previous work has attributed the contrast between quantification-compatible and quantification-resistant predicates to the lexical semantics of the predicate. Löbner (2000, 236) characterizes quantification-compatible predicates as summative: they hold of an individual just in case they hold of all its parts. Quantification-resistant

¹This phenomenon is sometimes referred to as subatomic quantification, where a quantifier ranges over parts of an entity that in other contexts can be conceptualized as atomic (i.e., lacking proper parts) or an integrated whole (Wagiel 2021). I avoid this term to prevent terminological confusion when introducing additional related notions later in the paper.

predicates, on the other hand, are integrative, meaning that they are true of an argument as an integral whole but do not necessarily apply to its parts.² According to this approach, *be narrow* and *be ruined* are summative, whereas *be short* and *tie the room together* are integrative.

In this paper, I argue that quantification compatibility is not a fixed lexical property of predicates. While some predicates tend to combine naturally with partitive quantifiers and others tend to resist them, these tendencies are not absolute. In appropriate contexts, even predicates that typically resist quantification over parts may permit it, and conversely. Ultimately, the acceptability of quantification over parts depends on the lexical semantics of the predicate, the part structure associated with the nominal argument, and the broader context.

To illustrate, consider the sentence *The answer is mostly correct*. Out of context, it sounds natural, suggesting that the predicate *be correct* is quantification-compatible. However, this compatibility is context dependent. Suppose the following exchanges occur during a school exam:

- (3) Teacher: What are the five largest U.S. states by area?
 Student: Alaska, Texas, California, Montana, and Kansas are the five largest U.S. states by area.
 Teacher: The answer is mostly correct, but the fifth-largest state is New Mexico, not Kansas.
- (4) Teacher: What is the largest U.S. state by area?
 Student: Texas is the largest U.S. state by area.
 Teacher: #The answer is mostly correct, but the largest state is Alaska, not Texas.

In (3), the student correctly names four out of five states, so the teacher can naturally say that the answer was mostly correct. In (4), by contrast, the student's answer lacks proper parts that could be independently correct or incorrect with respect to the teacher's question—it is simply right or wrong as a whole. As a result, the student's answer can be described as *almost* correct, but it would be unnatural to say that it was *mostly* correct. This contrast shows that quantification compatibility is not determined by the predicate alone, but depends on whether the context makes available a relevant part structure to which the predicate can meaningfully apply.³

Importantly, infelicity may also arise even when the predicate is well-defined for proper parts of the entity. Consider (5), said of a roughly round island. The predicate *straddle the equator* is true of regions whose spatial extent lies on both sides of the equator, and false of regions located entirely within one hemisphere. It is therefore defined for any spatial subpart of the island. Nonetheless, both (5-b) and (5-c) are infelicitous. Similarly, Istanbul straddles the boundary between Europe and Asia, but it is unnatural to say that it partly or entirely straddles that boundary.

- (5) a. The island straddles the equator.

²Corblin (2008) proposes a more fine-grained taxonomy, which includes predicates requiring a property to hold of most or at least some parts, rather than all.

³Löbner (2000, 237) argues that in such cases, the predication is effectively integrative.

- b. #Part of the island straddles the equator.
- c. #All of the island straddles the equator.

To capture the conditions under which quantification over parts of singular individuals is felicitous or infelicitous, I propose the following constraint:

- (6) *A constraint on quantification over parts* (CQP)
 Let P be the set of mereological parts of a singular individual x , let Q be a predicate, and let D be a partitive quantifier. A sentence ϕ of the form $D(P)(Q)$ is infelicitous if all of its scalar alternatives are contextually equivalent to it.

This constraint can be seen as a special case of a more general pragmatic principle governing sets of alternatives, and in particular scalar alternatives. The idea is that scalar alternatives compete with each other, and for this competition to be meaningful, they must be contextually distinct (Spector 2007, 2014; Magri 2009, 2011; Anvari 2018; Bar-Lev and Fox 2023; Bar-Lev 2024). This principle is stated as follows:⁴

- (7) *No trivial set of alternatives!* (NTA)
 If ϕ is associated with a set of alternatives ALT in a context C , then at least one member of ALT must be neither a C -contradiction nor a C -tautology, nor C -equivalent to ϕ . (Spector 2014, 9)

This principle has been invoked to account for the oddness of sentences such as (8). Although the two sentences are logically distinct (i.e., (8-a) makes an existential statement, while (8-b) makes a universal statement), they are contextually equivalent. Given that twins have the same age, the truth of (8-a) entails the truth of (8-b). In other words, the strengthened meaning in (9) is a contextual contradiction. As a result, neither alternative can be felicitously used, since the required contrast between the alternatives is absent.

- (8) *Context: Luise and Lottie are twins.*
 a. #One of the twins is nine years old.
 b. #Both of the twins are nine years old.
 (9) #One, but not both, of the twins is nine years old.

Turning to quantification over parts of singular individuals, I assume the following denotations for the non-distributive quantifiers *part* and *all*, where P is the set of mereological parts of the entity, Q is the predicate, and ' $\oplus$ ' represents a generalized sum operation:⁵

⁴CQP isolates the contextual-equivalence case of the more general NTA principle in (7). In the cases considered here, the relevant alternatives are neither contextual tautologies nor contradictions, so contextual equivalence is the operative condition.

⁵See Champollion and Krifka (2016) for an overview of mereological concepts and their applications in semantics.

The lexical entry for *all* is equivalent to assigning the determiner the denotation $\lambda P. \oplus P$, which then combines with the predicate by functional application. It thus resembles sum-based analyses of definite plurals and mass nouns (Sharvy 1980). For instance, this predicts *The water evaporated* and *All the water evaporated* to be equivalent. I do not assume, however, that *all* and definites are semantically identical. In particular, definites exhibit homogeneity and non-maximality effects that are absent with *all*. I leave the source of this contrast open, as discussed in Subsection 7.1. More importantly for the present analysis, *all* and *the* are associated with different alternatives: *all* competes with other quantifiers, whereas the

$$(10) \quad \llbracket \text{part} \rrbracket = \lambda P \langle e, t \rangle. \lambda Q \langle e, t \rangle. \exists y [P(y) \wedge Q(y)]$$

$$(11) \quad \llbracket \text{all} \rrbracket = \lambda P \langle e, t \rangle. \lambda Q \langle e, t \rangle. Q(\bigoplus P) \text{ (modified from Link 1983)}$$

Under these denotations, (5-b) means that some part of the island straddles the equator, while (5-c) means that the sum of all parts of the island (that is, the island as a whole) straddles the equator. Crucially, *straddle the equator* is upward monotonic with respect to parthood: if some part of the island straddles the equator, then the island as a whole does as well. The converse entailment also holds: if the island as a whole straddles the equator, then some part of it does, since the island itself is among its parts. The existential alternative (5-b) and the logically stronger universal alternative (5-c) are therefore contextually equivalent. Hence, these sentences violate NTA and are infelicitous, in parallel with the pattern in (8). Importantly, although quantification compatibility is closely tied to the lexical semantics of the predicate, it need not itself be lexically encoded, since it arises pragmatically.

A different way in which NTA can be violated is illustrated by the contrast between (3) and (4). In this case, the crucial factor is not the lexical semantics of the predicate, but whether the context makes available a suitable part structure.

In (3), the teacher’s question denotes a Hamblin set of possible answers, each of which may be correct or incorrect. Partial answers, such as “Texas is one of the five largest U.S. states by area”, are also defined for correctness, though with respect to subquestions like “Is Texas one of the five largest U.S. states by area?” rather than the superquestion. Thus, the student’s answer can be analyzed as a plurality of propositions (Schmitt Schmitt): a complete (though not fully accurate) answer to the superquestion, consisting of five partial answers to its subquestions, four of which are true. The teacher’s response in (3) is then interpreted as saying that most of these partial answers are correct.

By contrast, in (4), the predicate is defined only for the maximal element in P (the set of parts of the answer), namely the answer as a whole; proper parts fall outside the domain of entities to which *correct* can meaningfully apply in the given context. The student’s answer is incorrect both with respect to the superquestion and to the subquestion “Is Texas the largest U.S. state by area?” There is no further decomposition of the answer into parts that could independently count as correct or incorrect in the given context.

Crucially, this situation also leads to a violation of NTA. This is illustrated by (12): the logically weaker alternative in (12-a) contextually entails the logically stronger alternative in (12-b). If some part of the answer is correct, then the answer as a whole (the maximal element in P) must be correct, making (12-b) true as well.⁶ As a result, the scalar alternatives are contextually equivalent, and NTA is violated.

relevant alternatives to definites include indefinites. Crucially, when a definite and its indefinite alternative are contextually equivalent, the definite is favored by Maximize Presupposition (Heim 1991), as discussed in Subsection 6.2.

Another issue is whether the entry for *all* should use a maximality operator rather than a sum operator. In the cases considered here, P is the set of mereological parts of an entity, which is closed under sum and has the entity itself as its unique maximal element. Thus, $\bigoplus P$ coincides with the maximal element (Champollion and Krifka 2016, 384). The choice between a sum-based and a maximality-based entry therefore makes no difference for the cases discussed in this paper. It would matter, however, for restrictors that are not cumulative, as in *all ten students*. I thank two anonymous reviewers for raising these issues.

⁶There are two possible ways of construing the part structure of the answer. On one option, the answer is treated as atomic, in which case its set of parts is a singleton containing the answer itself. On the other

- (12) Teacher: What is the largest U.S. state by area?
 Student: Alaska is the largest U.S. state by area.
 Teacher:
 a. #Part of the answer is correct.
 b. #All of the answer is correct.

More generally, this pattern arises when all non-maximal elements in P are undefined for Q . In such cases, no proper part of the entity whose parts serve as the restrictor of the quantifier falls within the domain of the predicate, and the collapse of the distinction between existential and universal quantification (as well as non-endpoint scalar alternatives like *most*) leads to a violation of NTA.

The remainder of this paper is structured as follows. Section 2 outlines the empirical domain of the study. Section 3 reviews previous analyses of the distinction between quantification-compatible and quantification-resistant predicates, including in collective predication. Section 4 introduces the analysis of quantification over parts of singular individuals. Section 5 presents the main proposal and derives the distinction between quantification-compatible and quantification-resistant predicates. Section 6 extends the analysis to collective predication and object mass nouns. Section 7 discusses remaining issues. Section 8 concludes. The appendix compares the proposal with Winter (2002a) and argues that the proposed analysis does not hinge on the particular approach to non-distributive quantification adopted here.

2 Empirical domain

This paper focuses on quantification over parts of singular individuals. However, as noted by Löbner (2000) and Corblin (2008), the distinction between quantification-compatible and quantification-resistant predicates mirrors a similar phenomenon in the domain of collective predicates, known as the *gather/numerous* distinction (Kroch 1979; Dowty 1987; Taub 1989; Brisson 2003; Champollion 2010, 2015, 2017; Dobrovie-Sorin 2014; Kuhn 2020, among others). *Gather*-type collective predicates are compatible with plural quantification (13), whereas *numerous*-type collective predicates resist it (14).

- (13) a. All of the kids gathered outside.
 b. Most of the kids get along.
- (14) a. #All of the kids were numerous.
 b. #Most of the kids elected Mary as class president.

Dobrovie-Sorin (2014) further notes that a similar distinction arises in the domain of mass quantification and has been discussed under the heading of the Homogeneity Constraint (Roeper 1983; Lønning 1987; Higginbotham 1994):

- (15) a. Most gold is yellow.
 b. #Most gold weighs less than a ton.

option, the answer is decomposed into smaller parts, but these are not defined for the predicate. In either case, the distinction between existential and universal quantification collapses.

Although plural and mass quantification are not the primary focus of this paper, the parallels across the three domains suggest that they may admit a unified explanation. Accordingly, I occasionally extend the discussion beyond quantification over parts of singular individuals, particularly in reviewing previous studies (Section 3) and in Section 6, where the proposed analysis is extended to collective predication and object mass nouns.

The distinction between quantification-compatible and quantification-resistant predicates is observed with a range of determiners that quantify over parts or sub-pluralities of a whole, including *all*, *most*, *part*, *none*, and percentages (16). The same contrast also arises with cardinal partitives (17), although these are not discussed further here. For convenience, I will refer to this broader class of expressions as PARTITIVE QUANTIFIERS.

- (16) a. # {All / Most / Part / 80%} of the room is large.
b. # {All / Most / Some / None} of the kids were numerous.
- (17) a. #Four square meters of the room are large (and the rest is not).
b. #Forty of the kids were numerous (and the others were not).

Syntactically, the distinction between quantification-compatible and quantification-resistant predicates is evident across both determiners (18) and adverbial quantifiers (19). Examples in this paper alternate between these two types of quantifiers, selecting whichever form is most natural in a given context. A predicate will be considered quantification-compatible if it permits at least one of these quantificational expressions.⁷

- (18) a. All of the kitchen is {new / #large}.
- b. Most of the kitchen is {new / #large}.
- c. Part of the kitchen is {new / #large}.
- (19) a. The kitchen is entirely {new / #large}.
- b. The kitchen is mostly {new / #large}.
- c. The kitchen is partly {new / #large}.

⁷Adverbial quantifiers are sometimes ambiguous between quantification over parts, degree modification, and temporal modification (Aldridge and Neeleman 2015; Sudo 2025). When modifying gradable predicates, an adverb can be interpreted as a degree modifier (quality reading) rather than as quantifying over parts (quantity reading), as illustrated below:

- (i) The door is half transparent. (Sudo 2025)
 - a. Quantity reading: Half of the door is transparent.
 - b. Quality reading: The degree of transparency of the entire door is 50%.

Additionally, some adverbs permit a temporal (occasion) reading:

- (ii) The town is mostly quiet.
 - a. Quantity reading: Most of the town is quiet.
 - b. Occasion reading: The town is usually quiet.

This paper is concerned only with quantification over parts (i.e., quantity readings).

3 Previous studies

This section reviews the literature on the *gather/numerous* distinction in collective predication and examines whether and how these accounts can be extended to quantification over parts of singular individuals. Subsection 3.1 reviews the dominant approach to quantification compatibility, which ties it to the mereological property of divisive reference or the weaker notion of stratified reference.⁸ Subsection 3.2 presents three empirical challenges for extending the divisive/stratified reference approach to quantification over parts of singular individuals. Finally, Subsection 3.3 outlines the main respects in which the present analysis departs from previous work.

3.1 The divisive/stratified reference approach

A central idea in previous work on the *gather/numerous* distinction is that *gather*-type predicates are true of a plurality by virtue of being true of its subpluralities, whereas *numerous*-type predicates apply to the plurality as an integral whole (Dowty 1987; Löbner 2000). For example, (20) is true only if it holds of all subpluralities of the kids. In contrast, (21) is true if the group as a whole is large enough, with no requirement that its subpluralities satisfy the predicate.

(20) The kids gathered outside.

(21) The kids were numerous.

On this view, *gather*-type predicates are quasi-distributive: their distributive inferences apply to subpluralities (i.e., sums of individuals), rather than atomic individuals. By contrast, *numerous*-type predicates apply to the maximal plurality and do not distribute to its subparts. This is analogous to sentences like *The kid is laughing*, where the predicate applies to the individual as a whole and does not distribute to its parts (e.g., body parts).

This intuition has been formalized in terms of a weak form of divisive reference (Löbner 2000; Champollion 2010, 2015, 2017; Dobrovie-Sorin 2014; Kuhn 2020). Intuitively, a predicate is divisive if, whenever it is true of an entity, it is also true of all its parts. The formal definition is given below, where ‘ $\sqsubseteq$ ’ represents the parthood relation:

(22) *Divisive reference* (Champollion and Krifka 2016, 379)

A predicate P has divisive reference if and only if:

$$\forall x[P(x) \rightarrow \forall y[y \sqsubseteq x \rightarrow P(y)]]$$

A well-known challenge for divisive reference is the minimal-parts problem: certain collective predicates, such as *gather*, do not apply at the level of atomic individuals. For example, while a group of children may have gathered, it is not true of each individual child that they gathered. One way of accommodating this problem is to restrict divisivity to parts above a certain granularity (Winter 2002b). Thus, if a plurality of individuals gathered, sufficiently large subpluralities may also count as

⁸While most previous studies focus on the mereological properties that characterize *gather*-type and *numerous*-type predicates, Winter (2002a) is primarily concerned with their compatibility with quantification. I postpone the comparison between Winter’s approach and the present proposal to the appendix so as not to interrupt the flow of this section, which focuses on the divisive/stratified reference approach.

having gathered, while smaller subpluralities (in particular, atomic individuals) need not satisfy the predicate. Similarly, Löbner (2000, 236–237) argues that divisivity should be evaluated only with respect to parts for which the predicate is defined, as determined by its selectional restrictions. For instance, *The yolk is liquid* may be true even though the atoms making up its molecules are not themselves liquid.

However, Kuhn (2020) points out that restricting divisive reference by granularity is still too strong, an issue he terms the tricky-parts problem. Certain predicates such as *hold hands* behave as *gather*-type predicates, yet they need not apply to every sufficiently large subplurality. For example, *All the students held hands* is acceptable, even though it does not follow that every subgroup of students satisfies *hold hands*: typically, each student is only holding hands with one or two others. Thus, merely excluding parts below a certain granularity does not solve the problem, and *hold hands* would incorrectly be excluded from the *gather*-type class under this criterion.

To address this undergeneration problem, previous work appeals to stratified reference, a further weakening of divisive reference introduced by Champollion (2010) and subsequently developed by Champollion (2015, 2017). Intuitively, a predicate has stratified reference if entities in its extension can be decomposed into small parts that also satisfy the predicate. Crucially, stratified reference does not require the predicate to hold of every sufficiently large subpart, but only that the entity can be covered by small subparts that satisfy the predicate.

In Champollion’s analysis, the compatibility of *all* with collective predicates is constrained by stratified reference; that is, the quantifier presupposes that the predicate in its scope has stratified reference (Champollion 2010, 2015, 2017). Kuhn (2020) refines the application of this analysis to *gather*-type predicates by requiring the relevant strata to be small subpluralities. In Kuhn’s formulation, stratified reference is stated in terms of a cover: a set of possibly overlapping subpluralities whose mereological sum equals the whole. The granularity of the cover is determined by ϵ , a contextually determined small numerical value: each subplurality in the cover must have cardinality at most ϵ .⁹ Formally:

- (23) *Stratified reference* (Kuhn 2020, 232)
 A predicate P has stratified reference if and only if:
 $\forall x[P(x) \rightarrow \exists S[\bigoplus S = x \wedge \forall y \in S[|y| \leq \epsilon \wedge P(y)]]]$

Gather-type predicates such as *gather* and *hold hands* are correctly predicted to be admissible: any gathering plurality can be decomposed into small subpluralities of two or more participants who gathered, and any plurality of people holding hands can be divided into overlapping pairs of individuals holding hands. By contrast, *numerous*-type predicates lack stratified reference. For example, *be numerous* cannot hold of small subpluralities, since such groups are, by definition, not numerous. Likewise, a predicate like *return a verdict of not guilty* lacks stratified reference, as legal definitions

⁹Champollion (2017) provides a more general formulation of stratified reference, relativized to a dimension and a granularity parameter. In his application to the *gather/numerous* distinction, stratified reference is formulated over event predicates: a predicate like *gather* satisfies the relevant condition if any gathering event can be divided into one or more gathering subevents whose agents are sufficiently small pluralities. For presentational purposes, I adopt the simpler cover-based formulation in Kuhn (2020), defined directly over predicates of individuals.

prevent proper subpluralities (i.e., fewer than a full jury) from independently returning verdicts.

3.2 Empirical challenges

The stratified reference approach can in principle be extended to quantification over parts of singular individuals. Consider, for example, the sentence *The trail is narrow*. The predicate *be narrow* has stratified reference, since a narrow trail can be divided into smaller parts that are themselves narrow. This correctly predicts that the predicate is quantification-compatible, as illustrated in (1-a). However, as this subsection shows, this analysis faces three empirical challenges.

The first challenge concerns the behavior of gradable predicates. Consider the contrast in (24): *be high* (24-a) and *be wide* (24-b) are quantification-compatible, while *be long* (24-c) is quantification-resistant in this context. Importantly, all three predicates measure spatial extent, along different spatial dimensions.

- (24) a. Most of the Great Wall of China is 6–7 m high.
 b. Most of the Great Wall of China is 4–5 m wide.
 c. #Most of the Great Wall of China is 12,000 km long.

Moreover, the infelicity of *be long* in this context does not depend on the monotonicity of the measure phrase, as illustrated in (25). Importantly, a downward monotonic measure phrase such as *less than 12,000 km long* ensures that the property distributes from the whole to its parts: if an entity is less than 12,000 km long, then any subpart of it must also be shorter than 12,000 km. Therefore, under the stratified reference approach, such a predicate is expected to be quantification-compatible. However, this prediction is not borne out.¹⁰

- (25) #Most of the Great Wall of China is {exactly / more than / less than} 12,000 km long.

¹⁰This problem has already been noted by Champollion (2017) and Kuhn (2020) in the context of collective predicates. With a downward monotonic predicate such as *be few in number*, any subplurality of a group that is few in number is also few in number. As a result, the stratified reference approach predicts that this predicate should behave like a *gather*-type predicate and thus allow plural quantification, contrary to fact, as shown in (i):

- (i) #All of the kids were few in number.

As noted by Champollion (2017, 248), this problem extends to analyses of other constructions that have been argued to require divisive or stratified reference, such as *for*-adverbials. For example, (ii) is unacceptable despite the fact that the event can be decomposed into short subevents of drinking less than a glass of wine.

- (ii) #Bill drank less than a glass of wine for an hour.

To address this problem in the context of collective predicates, Kuhn (2020) refines the notion of stratified reference by introducing the concept of grounded stratified reference. The key insight is that any plurality can be divided into smaller subpluralities that are trivially few in number, regardless of how large the whole group actually is. Consequently, although *be few in number* technically satisfies stratified reference, this decomposition is not genuinely tied to the property of the entire plurality. By requiring that stratified reference be grounded in the property of the whole, Kuhn correctly predicts that *be few in number* should align with *numerous*-type predicates. See also Champollion et al. (2017) for a different account of this problem, particularly in the context of *for*-adverbials.

In Subsection 5.3, I offer an alternative analysis of the distinction between quantification-compatible and quantification-resistant gradable predicates, which attributes the infelicity of sentences like (25) and (i) to a violation of NTA (7).

The second challenge concerns existential predicates such as *be inhabited*, which are quantification-compatible, as seen in (26), despite lacking stratified reference. To see why the predicate itself lacks stratified reference, consider *The island is inhabited*. The sentence can be true even if only a small region of the island is populated. It does not entail that every region of the island is inhabited, nor that the island can be divided into small inhabited regions (unless one resorts to creative gerrymandering). Despite lacking stratified reference in this sense, the predicate remains compatible with quantification over parts of a singular individual.¹¹

(26) Most of the island is inhabited.

The third challenge concerns the observation that singular count nouns such as *an island* can function as quantification-compatible nominal predicates, as shown in (27).¹² This is unexpected, since no proper part of an island is itself an island. Despite the absence of stratified reference, the predicate can nevertheless combine naturally with a partitive quantifier such as *partly*.

(27) Mombasa is partly an island.

3.3 Departures from previous approaches

The analysis developed here departs from previous accounts in several respects. First, it focuses primarily on quantification over parts of singular individuals, rather than on plural quantification with collective predicates. However, I extend the analysis to collective predicates in Section 6.

Second, earlier studies have typically started from sentences with definite subjects, such as *The trail is {narrow / short}*. The usual strategy has been to examine whether such sentences give rise to quasi-distributive inferences and, based on their presence or absence, to predict whether the predicate is quantification-compatible (Dowty 1987;

¹¹An anonymous reviewer points out that the acceptability of (26) can be accounted for under the stratified reference approach by appealing to meaning shift. According to Champollion (2017, 255–256), when a predicate lacks stratified reference, its meaning can sometimes be shifted via a distributivity operator. This helps explain the observation in Kroch (1979) that *be numerous* is compatible with *all* when the restrictor is a group noun, as in:

(i) All the enemy armies were numerous.

In this case, the sentence has a distributive reading, in which each enemy army has many soldiers; a collective reading, where the number of armies is large, is unavailable. Importantly, once the predicate is shifted by the distributivity operator, it acquires stratified reference (Champollion 2017, 255–258).

By the same reasoning, applying the distributivity operator to *be inhabited* yields a distributive predicate, allowing *most* in (26) to quantify over cells of a contextually salient cover. However, this analysis requires an independent account of what determines the relevant cover, an issue I address in Subsection 4.2.

¹²Mombasa, Kenya, is centered around Mombasa Island but extends onto the mainland. A similar naturally occurring example is given below:

(i) Kennedy Space Center [...] occupies what is partly an island, Merritt Island, and partly a peninsula off the coast of Florida. [Blanton McBride, “You Can Have a Blast Seeing Apollo 15 Off”, *New York Times*, 18 July 1971, <https://www.nytimes.com/1971/07/18/archives/you-can-have-a-blast-seeing-apollo-15-off-an-angler-hooked-a-trout.html>]

A further example with a different nominal predicate is:

(ii) In other words, most of a snake’s skeleton is an extremely extended thorax. [“Snake”, *Wikipedia*, accessed 29 July 2026, <https://en.wikipedia.org/wiki/Snake>]

Löbner 2000). By contrast, the present analysis begins directly with quantificational sentences like *Most of the trail is {narrow / #short}*. Given this point of departure, the account does not assume, contrary to the intuition widely shared since Dowty (1987) (cf. Winter 2002a), that quantification compatibility must be linked to inferences from the whole to the parts. As shown in Subsection 3.2, some quantification-compatible predicates do not give rise to quasi-distributive inferences, which reinforces the value of this more direct approach.

Finally, similarly to Winter (2002a), this paper focuses on explaining quantification compatibility rather than on identifying a single mereological property that characterizes quantification-compatible predicates. Unlike Winter’s account, however, I do not assume a lexical distinction between atom and set predicates (see the appendix for a comparison with Winter 2002a). Instead, the proposed account relies on independently motivated lexical properties of predicates, such as additivity of measure functions (see Subsection 5.3).

4 Quantification over parts of singular individuals

4.1 Preliminary version

Quantification over parts of singular individuals is a type of non-distributive quantification. I assume that non-distributive determiners denote binary relations between sets, in parallel with the relational view of distributive quantifiers (Barwise and Cooper 1981).¹³

$$(28) \quad \llbracket \text{part} \rrbracket = \lambda P \langle e, t \rangle. \lambda Q \langle e, t \rangle. \exists y [P(y) \wedge Q(y)]$$

$$(29) \quad \llbracket \text{all} \rrbracket = \lambda P \langle e, t \rangle. \lambda Q \langle e, t \rangle. Q(\bigoplus P)$$

$$(30) \quad \llbracket \text{most} \rrbracket = \lambda P \langle e, t \rangle. \lambda Q \langle e, t \rangle. \exists y [P(y) \wedge Q(y) \wedge \mu(y) > \frac{\mu(\bigoplus P)}{2}]^{14}$$

¹³Non-distributive *all* has been analyzed in different ways: as having a distinct lexical entry from the singular count quantifier *every* (Link 1983), as synonymous with *every* but involving type shifting (Scha 1981; van der Does 1993; Winter 2002a,b), or as having the same semantics as distributive *every* (Haslinger et al. 2025). In this paper, I treat *all* as semantically distinct from *every*, while noting that the same truth conditions could in principle be derived under alternative analyses, as detailed in the appendix. Additionally, note that *part* can also function as a count noun. This use is distinct from the use of *part* as a determiner, which is distinguished by lacking an indefinite article or determiner, since English normally does not allow bare singular count nouns.

(i) Part of the island is in the Northern Hemisphere. (determiner)

(ii) Some part of the island is in the Northern Hemisphere. (count noun)

The same distinction applies to *half*, which can function as either a determiner or a count noun:

(iii) Half of the island is in the Northern Hemisphere. (determiner)

(iv) One half of the island is in the Northern Hemisphere. (count noun)

See Wągiel (2021) for an analysis of *part* and *half*, as well as other quantificational expressions, as count nouns.

¹⁴This denotation does not require a single connected region satisfying the predicate to exceed half the measure of the whole. Since the parthood relation is closed under sum, *y* may consist of several disjoint parts. As developed below, the obligatory application of the star operator at the VP level allows the predicate to hold of such a sum when it holds of its constituent parts.

In quantification over parts of a singular individual, the restrictor of the quantifier is the set of mereological parts of the nominal argument. Specifically, this set is introduced by a partitive marker, which maps entities to sets of their mereological parts (type $\langle e, \langle e, t \rangle \rangle$). Formally, the partitive marker encodes the parthood relation between entities:¹⁵

$$(31) \quad \llbracket \text{PARTITIVE} \rrbracket = \lambda y \lambda x. x \sqsubseteq y$$

The partitive marker may be overt, as in the English preposition *of* (Barker 1998), or covert. For example, $\llbracket \text{of the island} \rrbracket$ denotes the set of parts of the island.

The parthood relation is standardly taken to be closed under sum, meaning that the sum of any two parts of an entity is itself a part of that entity. Thus, the set of mereological parts of an entity forms a join-semilattice, similarly to the extensions of mass nouns and pluralized count nouns (Link 1983).

According to Ionin et al. (2006), the partitive marker is underspecified with respect to the type of parthood relation, which is determined by its complement. Following Link (1983), two types of parthood relations are typically distinguished: an individual parthood relation ($\sqsubseteq_i$), holding between individuals and sums thereof (i.e., pluralities), and a material parthood relation ($\sqsubseteq_m$), holding between portions of matter. For instance, in *Most of the rugs are on sale*, the partitive marker is interpreted in terms of individual parthood, so *of the rugs* denotes subpluralities of the rugs, including individual rugs but not their material parts. By contrast, in *Most of the rug is ruined*, the partitive marker is interpreted in terms of material parthood, so *of the rug* denotes the set of material parts of the rug.¹⁶

The distinction between individual and material parthood is needed to prevent plural quantification from ranging over material parts of singular individuals within the extension of a plural NP. For example, (32) is not judged true if only a small part of one of the rugs is red. Since the partitive marker is interpreted in terms of individual parthood, the restrictor consists of individual rugs and pluralities formed from them. Consequently, the presence of a red part within a rug does not suffice to make the sentence true.

$$(32) \quad \text{Some of the rugs are red.}$$

At the same time, the relevant contrast is not simply between individual parthood ($\sqsubseteq_i$) and material parthood ($\sqsubseteq_m$), since quantification over parts of singular individuals need not be restricted to material constituents. For instance, (33-a) concerns the informational content of a book, while (33-b) plausibly ranges over components of a complex aggregate measure.

¹⁵Barker (1998) argues that partitives encode the proper parthood relation $\sqsubset$ instead of $\sqsubseteq$. I follow Ionin et al. (2006) in allowing improper parthood.

¹⁶An alternative view posits that the partitive marker cannot directly combine with an atomic individual (Ladusaw 1982; Barker 1998). This is predicated on an ontological distinction between a count domain and a mass domain, which are disjoint but are connected through a relation of constituency (Link 1983). Under this view, a sentence like *Most of the rug is ruined* requires an operator that maps the individual rug, which is a member of the count domain, to its material parts in the mass domain.

Yet another approach assumes a unified parthood relation that applies to both individual parts and material parts (Moltmann 1997, 1998; Wągiel 2021). However, this raises the problem that partitives could in principle refer to material parts of individuals within a plurality. To avoid this, Wągiel (2021, 243-244) proposes restricting the transitivity of the parthood relation.

- (33) a. Most of the book is in French.
b. Most of the recent inflation is due to supply-chain disruptions.

Importantly, I do not assume multiple primitive notions of non-individual parthood. Rather, I posit a single non-individual parthood relation $\sqsubseteq_p$. Different kinds of entities may nevertheless come with different part structures. For example, a physical book has material parts such as leaves and ink, whereas its informational content has parts such as chapters, paragraphs, and sentences.

Against this background, consider the analysis of existential quantification over parts using the determiner *part* (28), as in (34): the sentence is true just in case some part of the island is located in the Northern Hemisphere.

- (34) Part of the island is in the Northern Hemisphere.

In many cases, universal quantification over parts of singular individuals can be analyzed in parallel with distributive quantification: each part of the entity independently satisfies the predicate, i.e., $\forall y[P(y) \rightarrow Q(y)]$. On this distributive analysis, (35) is true just in case every relevant part of the island is located in the Northern Hemisphere.

- (35) All of the island is in the Northern Hemisphere.

However, this analysis does not yield the correct truth conditions for sentences like (36), where the predicate holds only of the maximal element in P . The sentence does not mean that each part of the land west of the fault line became an island independently. Rather, it means that the entire area was cut off by the sea and thereby formed a single island. If *all* were analyzed as $\forall y[P(y) \rightarrow Q(y)]$, (36) would incorrectly be predicted to require a distributive reading, even though it is not possible for every arbitrary part to be an island on its own.

- (36) Following the earthquake, all the land west of the fault line became an island.

By contrast, the non-distributive analysis of *all* in (29) yields the correct truth conditions: the sum of the land west of the fault line became an island.¹⁷

As for the proportional quantifier *most* (30), it introduces a measure function μ , which maps entities to degrees on a scale whose dimension is contextually determined, such as volume, area, or mass (Schwarzschild 2006). For example, (37) is true just in case there exists a part of the land west of the fault line that became an island whose measure along the relevant dimension exceeds half the measure of the total land. Under the most salient interpretation of (37), the relevant dimension is projected area, as seen in a two-dimensional map.

- (37) Following the earthquake, most of the land west of the fault line became an island.

Turning to adverbial quantifiers, I assume that while determiners are of type $\langle\langle e, t \rangle, \langle\langle e, t \rangle, t \rangle\rangle$, adverbial quantifiers are of type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$. Unlike determiners,

¹⁷The non-distributive analysis of *all* is further supported by plural collective predication. For example, *All the kids gathered outside* cannot mean that every subplurality of the kids gathered outside, since subpluralities consisting of a single individual cannot satisfy the predicate, which selects a plural argument.

where the parthood relation is introduced by a separate partitive marker, adverbial quantifiers encode the parthood relation within their lexical semantics.

$$(38) \quad \llbracket \text{partly} \rrbracket = \lambda Q \langle e, t \rangle. \lambda x. \exists y [y \sqsubseteq x \wedge Q(y)]$$

$$(39) \quad \llbracket \text{entirely} \rrbracket = \lambda Q \langle e, t \rangle. \lambda x. Q(x)$$

$$(40) \quad \llbracket \text{mostly} \rrbracket = \lambda Q \langle e, t \rangle. \lambda x. \exists y [y \sqsubseteq x \wedge Q(y) \wedge \mu(y) > \frac{\mu(x)}{2}]$$

For example, (41) is true just in case Nauru as a whole is an island.¹⁸ As in the case of (36), a distributive analysis that quantifies over all parts of the entity would not yield the correct truth conditions.

$$(41) \quad \text{Nauru is entirely an island.}$$

Finally, I abstract away from distributional restrictions related to the mass/count distinction. For instance, *much* is compatible with both mass quantification (e.g., *much water*) and quantification over parts of singular individuals (e.g., *much of the island*), whereas *many* requires a plural noun phrase (e.g., *many of the students*). Similarly, *part* and *some* are both existential but differ in their distribution: *part* is compatible with quantification over parts of singular individuals and mass nouns but not plurals, whereas *some* is compatible with plurals and mass nouns but not with singular individuals (Schwarzschild 1996). This contrast is illustrated by the following examples from Sudo (2025):

- (42) a. Part of the door is wooden.
b. Part of the water has evaporated.
c. *Part of the doors are wooden.

- (43) a. *Some of the door is wooden.
b. Some of the water has evaporated.
c. Some of the doors are wooden.

Sudo (2025) accounts for these patterns by assuming that these quantifiers lexically encode presuppositions regarding the domain of quantification. I set these distributional restrictions aside and treat pairs such as *part/some* and *much/many* as semantically equivalent for the purposes of the analysis.

4.2 Refining the analysis

The preliminary analysis proposed in the previous subsection fails to derive the correct truth conditions for sentences involving existential predicates, such as *be inhabited* in (44). For simplicity, assume that a region counts as inhabited if it has permanent residents; otherwise it is uninhabited. As an existential property, inhabitedness is upward monotonic with respect to parthood: if a region is inhabited, then any region containing it is also inhabited. Thus, a large island with only a tiny village can be

¹⁸Here is a similar naturally occurring example, referring to Prince Edward Island, a province of Canada:

- (i) This is the only province which is entirely an island. [web example: <https://umaine.edu/canam/learn-the-provinceterritories-of-canada/prince-edward-island/>]

said to be inhabited. Similarly, an intergalactic encyclopedia could truthfully state that the solar system, or even the Milky Way Galaxy, is inhabited, even if it turns out that only the Earth has residents.

(44) All of the island is inhabited.

The truth conditions of (44) do not follow straightforwardly from the lexical entry for *all* in (29). Suppose that there is just one person living on a large island. In that case, the island as a whole counts as inhabited, and the sentence is predicted to be true. However, these truth conditions are too weak: intuitively, (44) means that each relevant part of the island is inhabited independently, not merely that the island as a whole is inhabited. These stronger truth conditions can be derived under a distributive analysis of *all* (i.e., $\forall y[P(y) \rightarrow Q(y)]$) together with a partition of the island into subregions. However, as previously argued in Subsection 4.1, this analysis faces a problem in cases where only the maximal element in P satisfies Q , as in (36).

I propose that the truth conditions of (44) can be derived by assuming a cover variable that restricts the domain of quantification (Higginbotham 1981; Gillon 1987; Schwarzschild 1996), together with a star operator that renders the predicate cumulative (Link 1983), following the analysis of plural predication in Bar-Lev (2025b).

A cover of an entity x is a set whose sum is identical to x . Intuitively, a cover specifies a way of dividing x into parts, possibly overlapping ones. Accordingly, a cover of x is a subset of the set of mereological parts of x . Formally:

(45) *Cover*
A set S is a cover of an entity x if and only if: $\bigoplus S = x$.

I assume that a cover variable C introduces a contextually salient set Cov^c that covers the sum of the individuals in the domain of discourse ($\bigoplus D$) (Heim 1994; Beck 2001; Bar-Lev 2025b, following Schwarzschild 1993, 1996). The C head combines with the predicate by Predicate Modification. Thus, the cover restricts the domain to those entities that are members of the cover:

(46) $\llbracket \text{is inhabited} \rrbracket = \lambda x. \text{INHABITED}(x)$
(47) $\llbracket C \text{ is inhabited} \rrbracket^c = \lambda x. Cov^c(x) \wedge \text{INHABITED}(x)$

For example, a large island may itself be in the extension of the unmodified predicate *is inhabited*. However, if the cover is restricted to small regions, so that the island as a whole is not a member of Cov^c , then the island will not be in the extension of the cover-modified predicate.

Importantly, in quantification over parts, the cover restricts the nuclear scope of the quantifier, not its restrictor. Thus, in (44), the restrictor P is the set of all parts of the island, whereas the nuclear scope Q consists of inhabited regions that are also members of the cover. This means that merely restricting the cover to small subregions of the island is not enough to derive the intended interpretation, because *all* still applies the predicate to the sum of all parts, namely the island as a whole. Since the whole island is not itself one of the small regions in the cover, the sentence would come

out false, even if every subregion of the island included in the cover were inhabited. Put differently, without further machinery, the cover can restrict the predicate to small regions, but it cannot make the whole island satisfy the predicate in virtue of being composed of such regions.

Following Bar-Lev (2025b), I therefore assume that a star operator (Link 1983) applies to predicates modified by a cover variable. The role of the star operator is to make the modified predicate cumulative, so that a larger entity may satisfy it whenever it can be built up from smaller parts that satisfy it. Applied to a predicate Q , the star operator yields the closure of Q under sum:

$$(48) \quad \begin{aligned} &\textit{The star operator} \text{ (Champollion 2016, 24)} \\ &x \in {}^*Q =_{\text{DEF}} \exists S[x = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq Q] \\ &\text{(i.e., } x \text{ is the sum of all the elements of a non-empty subset of } Q\text{)} \end{aligned}$$

According to this definition, $x \in {}^*Q$ holds if and only if x is composed of one or more parts that satisfy Q . Equivalently, either x itself is in Q , or x can be decomposed into parts, each of which is in Q .

Applying the star operator to a predicate modified by a cover variable yields the following meaning:

$$(49) \quad \begin{aligned} &\llbracket {}^*[C \text{ is inhabited}] \rrbracket^c = \\ &\lambda x. \exists S[x = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq (Cov^c \cap \llbracket \text{is inhabited} \rrbracket)] \end{aligned}$$

Intuitively, this means that x is entirely composed of inhabited regions drawn from the contextually determined cover. For example, if the cover consists of 1 km² regions, then the universal statement in (44) is true if the island can be decomposed into such regions, each of which has residents.

Similarly, (50) means that there exists a region of the island whose measure exceeds half of that of the entire island (e.g., in terms of area), and such that this region consists of one or more parts, each of which is an inhabited region in the cover.

$$(50) \quad \text{Most of the island is inhabited.}$$

Building on analyses that take the star operator to apply obligatorily in plural predication (Schwarzschild 1993; Sauerland 1998; Kratzer 2008; Bar-Lev 2025b,a), I assume that the star operator obligatorily applies to VPs in both plural and singular predication. This departs from Kratzer (2008), who restricts the star operator to predicates that are sisters of plural DPs, and possibly from Bar-Lev (2025a), who assumes that the star operator obligatorily applies to every predicate except singular NPs, but leaves open whether it also applies to singular verbs and VPs.¹⁹

Both the cover variable and the star operator are motivated independently of overt quantification, at least in the case of existential predicates, by the interpretation of sentences with definite subjects. Consider (51-a) and (51-b). The default interpretation

¹⁹My assumption is broadly compatible with Wehbe (2023, 2026). Wehbe (2026, 45–49) argues that the star operator cannot apply directly to lexical verbs or VPs, but only to predicates derived by movement. Wehbe (2026, 86–90) considers the possibility that the star operator obligatorily applies to every such derived predicate, including those formed by movement of a singular DP. The distinction between applying the star operator directly to the VP and applying it to a predicate derived by movement is not crucial for the cases discussed here, and I therefore abstract away from the movement involved.

of (51-a) requires only that some part of the island be inhabited, regardless of whether other parts are uninhabited. However, (51-b) shows that a stronger reading is also available. The sentence implies that all relevant parts of the island are inhabited, except for the specified region.²⁰

- (51) a. The island is inhabited.
b. The island is inhabited, except for a rugged area in the west.

The availability of these two readings can be captured if the predicate is evaluated relative to a contextually determined cover and modified by a star operator. The cover determines which parts are relevant, while the star operator makes it possible for the whole island to satisfy the predicate in virtue of being composed of such parts. Thus, I assume the LF in (52).

- (52) The island is inhabited.
LF: [The island] [*[C [is inhabited]]]

If the cover contains the island as a whole, possibly alongside some proper parts, the sentence is true as long as some region of the island is inhabited, given the existential meaning of the unmodified predicate *is inhabited*. This choice of cover therefore yields the default existential reading in (51-a). By contrast, if the cover excludes the island as a whole and instead consists of smaller, contextually relevant regions (e.g., 1 km² cells), a single inhabited region is no longer sufficient. Instead, the island satisfies the starred predicate only if it can be built up from inhabited regions drawn from the cover. This yields the stronger reading observed in (51-b): the island must be exhaustively covered by inhabited regions drawn from the cover, modulo the region explicitly excluded by the exceptive phrase.

This brings us to the question of what determines the choice of cover. As argued in the literature on plural predication, it is not enough that there merely exists a cover under which a sentence is true. Rather, the cover must be contextually salient (Schwarzschild 1996, following Lasnik 1989).

Building on this idea, Bar-Lev (2025b) distinguishes three types of covers that are typically contextually accessible in plural predication: (i) power cover: includes all parts of the entity (effectively equivalent to having no cover at all); (ii) atomic cover: consists exclusively of the entity's atomic parts; and (iii) singleton cover: contains only the entity as a whole, without any of its proper parts.²¹ Each type of cover gives rise to a characteristic interpretation: an atomic cover yields a distributive reading, a singleton cover yields a collective reading, while a power cover leaves the interpretation underspecified between distributive and collective (termed a cumulative interpretation in Bar-Lev 2025a). Following Bar-Lev (2025b), the cover variable, as well as the star operator, are assumed to be always present in the structure; in cases where no domain restriction is intended, the variable is assigned a power cover.

Which type of cover is involved in quantification over parts of singular individuals with existential predicates, as in (44) and (50)? As previously argued, a power cover (equivalently, the absence of a cover restricting the domain of quantification) does

²⁰I thank an anonymous reviewer for suggesting this example.

²¹In Bar-Lev (2025a), these are called cumulative, distributive, and collective covers, respectively.

not yield the correct interpretation. Because the predicate is existential in nature, the island as a whole counts as inhabited as soon as some part of it is inhabited. Consequently, if the island as a whole is included in the cover, both (44) and (50) are predicted to be true, even if the area where people actually live covers only a small fraction of the island. A singleton cover, containing only the island as a whole, is inadequate for the same reason (more generally, it is incompatible with quantification over parts, as will be argued in Section 5).

Of the three cover types considered above, the remaining option is an atomic cover, which consists of the atomic parts of the island. But what counts as an atomic part of an island? Atoms are defined relative to a domain and a parthood relation. In quantification over parts, the relevant parthood relation is introduced by the partitive marker. I write this relation as $\sqsubseteq_c$ (and its strict counterpart $\sqsubset_c$), where $\sqsubseteq_c$ may be instantiated either as individual parthood $\sqsubseteq_i$ or as non-individual parthood $\sqsubseteq_p$ (see Subsection 4.1). An atom with respect to $\sqsubseteq_c$ is an entity that has no proper $\sqsubseteq_c$ -parts:

$$(53) \quad \text{Atom} \\ \text{atom}_c(x) =_{\text{DEF}} \neg \exists y [y \sqsubset_c x]$$

For example, in *Most of the rugs are on sale*, the partitive marker is interpreted in terms of individual parthood $\sqsubseteq_i$, given that the NP is plural (Ionin et al. 2006). On this construal, the atomic elements are the individual rugs: each singular rug is minimal with respect to $\sqsubseteq_i$, since it has no proper individual parts. This does not deny that a rug has material parts; rather, such parts fall under non-individual parthood $\sqsubseteq_p$ and therefore do not count as parts in the sense of $\sqsubseteq_i$.

By contrast, a non-individual parthood relation $\sqsubseteq_p$ does not, in general, guarantee the existence of atoms. Consider *Most of the rug is ruined*, where the domain of quantification is plausibly the material parts of the rug. According to Link (1983), the mass domain is atomless: any material part of the rug can be further divided into proper parts (cf. Chierchia 1998, 2010; Rothstein 2010, 2017). On this view, $\sqsubseteq_p$ does not induce a set of minimal elements in this case.²²

This creates a tension for the present analysis. On the one hand, as previously argued, (44) cannot be analyzed using a power cover or a singleton cover. On the other hand, an atomic cover is likewise unavailable if the domain of quantification consists of material parts of the island, since this domain is arguably atomless.

I propose that this tension can be resolved by allowing the nominal expression to receive a spatial interpretation. This possibility is independently motivated by sentences such as (54). This sentence cannot be analyzed as quantification over material parts of the solar system, since void is, by definition, a region of space devoid of matter. Rather, *the solar system* is understood here as referring to the region of space associated with the solar system, and the sentence states that most of this spatial region is void. Similarly, in (44), *the island* may be interpreted as referring to the spatial region associated with the island rather than to the concrete island itself.

²²In other cases, however, atoms may exist in the domain with respect to a non-individual parthood relation $\sqsubseteq_p$. For instance, in *Most of the book is in French*, the relevant domain is the book's informational content. Parts of the content must be linguistic expressions of some kind, yielding a domain that is naturally treated as discrete, and hence compatible with atoms, even if it remains open whether the most plausible atoms are words, morphemes, or other units.

(54) Most of the solar system is void.

To formalize this idea, I assume that the domain of individuals D_e contains spatial-region individuals in addition to concrete and other kinds of individuals. Spatially interpretable definite descriptions and proper names, such as *the island* or *France*, may denote either a concrete individual or a corresponding spatial-region individual, depending on the context. The distinction is useful because the part structure of a spatial-region individual need not coincide with that of the corresponding concrete individual. For example, a cross-section through the region occupied by a wall may be part of that region even if there is no independently individuated material part of the concrete wall corresponding to that cross-section.

Following work on the semantics of spatial adpositions (Wunderlich 1991; Zwarts and Winter 2000), I assume a domain of points D_p and a location function loc that maps spatially located individuals to their spatial representation, or eigenspace, modeled as a non-empty set of spatial points (type $\langle p, t \rangle$). I further assume that spatial-region individuals are individuated by their locations: if x and y are spatial-region individuals and $loc(x) = loc(y)$, then $x = y$. This assumption applies only to spatial-region individuals and therefore does not preclude a concrete individual and a spatial-region individual from having the same location. Thus, if i is a concrete island and s_i is the corresponding spatial-region individual, it may be the case that $i \neq s_i$ while $loc(i) = loc(s_i)$.

The non-individual parthood relation $\sqsubseteq_p$ also applies to spatial-region individuals. For such individuals, parthood corresponds to inclusion between their eigenspaces:

(55) *Parthood among spatial regions*

If x and y are spatial-region individuals, then $x \sqsubseteq_p y$ iff $loc(x) \subseteq loc(y)$.

Since subset inclusion is reflexive and transitive, and spatial-region individuals are individuated by their locations, $\sqsubseteq_p$ constitutes a partial order over spatial-region individuals. As before, the partitive marker introduces a contextually determined part-hood relation. When its complement denotes a spatial-region individual, the relevant relation is $\sqsubseteq_p$ as instantiated over spatial regions. Thus, if *the island* denotes the spatial-region individual s_i , *of the island* denotes the set of spatial-region individuals that are $\sqsubseteq_p$ -parts of s_i .

I further assume that for every spatial point $p \in D_p$, there is a spatial-region individual $x \in D_e$, which I call a point-region individual, such that $loc(x) = \{p\}$. The inclusion of such individuals in the domain of discourse is independently plausible: natural language can refer to point-like spatial regions using definite descriptions, such as *the center of the circle*, or by means of a coordinate system. Importantly, point-region individuals are atoms with respect to $\sqsubseteq_p$: since a singleton has no proper non-empty subsets, no spatial-region individual is a proper part of a point-region individual.

When *the island* in (44) denotes the corresponding spatial-region individual s_i , its point-region parts provide the relevant atomic cover. Their singleton locations jointly exhaust $loc(s_i)$, and any spatial-region individual that contains all of these point-regions as parts must therefore also contain s_i as a part. Hence, s_i is their least

upper bound under $\sqsubseteq_p$, and thus their sum. Since the point-region individuals are atoms, they therefore form an atomic cover of s_i . The spatial-region interpretation thus provides the atomic cover required by the analysis without implying that the concrete island itself is composed of points.

What remains is to define *is inhabited* for point-region individuals. Recall that the location of a point-region individual is a singleton containing a spatial point. Whether such an individual counts as inhabited can be determined by considering a small neighborhood around this point. Let r be a contextually determined parameter specifying the radius of a circular region centered on the point. A point-region individual counts as inhabited if there are inhabitants within this region. The resulting partition of the island's point-region parts into inhabited and uninhabited ones depends on the choice of r : a small radius yields a representation that is highly sensitive to fine-grained local variation, while a larger radius reflects broader regional trends. In this way, the analysis determines which point-region parts of the island count as inhabited and whether their sum covers most or all of the spatial-region individual corresponding to the island in terms of area.²³

In sum, the analysis of sentences involving quantification over parts of singular individuals sometimes requires a cover that restricts the parts relative to which the predicate is evaluated, as illustrated by *All of the island is inhabited*. In other cases, however, no such restriction is needed, and the relevant interpretation can be derived by considering all parts, as in *All of the island is in the Northern Hemisphere*. For the sake of uniformity, I will nonetheless assume that a cover variable is always associated with the predicate, with a power cover corresponding to cases in which the set of relevant parts is effectively unrestricted.

5 Deriving quantification compatibility

Building on the analysis of quantification over parts of singular individuals developed in the previous section, this section turns to deriving quantification compatibility. I begin by restating the constraint proposed above, repeated here as (56). This constraint can be seen as a special case of NTA (7).

- (56) *A constraint on quantification over parts (CQP)*
 Let P be the set of mereological parts of a singular individual x , let Q be a predicate, and let D be a partitive quantifier. A sentence ϕ of the

²³Whether a point-region individual can possess a property like being inhabited is arguably a metaphysical question. However, I see no linguistic reason to rule it out. Point-like regions rarely function as overt arguments in natural language, so there is little direct evidence that predicates like *be inhabited* select for regions of non-zero size. This contrasts with collective predicates like *meet*, which do select for plural arguments, hence the unacceptability of sentences such as **John met yesterday*. In principle, the same truth conditions could also be derived using a pixel-based cover—for example, by partitioning the island into 1-km² pixels, where the pixel size plays a role analogous to the r parameter. However, such covers introduce unnecessary complications. An atomic cover is also well-suited (though not strictly required) for predicates such as *is in the Northern Hemisphere*, which are naturally defined pointwise. A pixel-based cover, on the other hand, imposes arbitrary quantization and also creates technical problems: pixels that straddle the equator would have to be excluded, since they do not clearly fall within either the positive or the negative extension of the predicate. Although these issues could potentially be resolved, it is preferable to avoid them altogether by adopting an atomic cover or, when appropriate, a power cover. I will therefore not make use of pixel-based covers in the analysis developed here.

form $D(P)(Q)$ is infelicitous if all of its scalar alternatives are contextually equivalent to it.

To determine whether a set of alternatives violates CQP, it is sufficient to consider whether the weakest and strongest alternatives in the set of alternatives are contextually equivalent. In upward-entailing environments, these correspond to existential and universal quantification, respectively. Other scalar alternatives—such as those formed with *much* or *most*—occupy intermediate positions on the same scale. If the weakest alternative (e.g., a sentence with the existential quantifier *part*) is contextually equivalent to the strongest alternative (e.g., the corresponding sentence with the universal quantifier *all*), then all intermediate alternatives are necessarily contextually equivalent as well. This follows because the alternatives are partially ordered by strength: each intermediate alternative entails the weakest one and is entailed by the strongest one.²⁴

Consider the following scalar alternatives, where DP denotes a singular individual, such as a definite description like *the island* or a proper name like *Iceland*:

- (57) a. Part of DP VP .
b. All of DP VP .

Recall that a cover variable C introduces a contextually determined cover Cov^c that restricts the extension of the predicate to members of the cover, and that a star operator makes the modified predicate cumulative (see Subsection 4.2). I assume that scalar alternatives are evaluated relative to the same value of the cover variable. Thus, replacing the quantifier does not alter the contextually supplied cover.

$$(58) \quad \llbracket * [C \text{ VP}] \rrbracket^c = \lambda x. \exists S [x = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq (Cov^c \cap \llbracket VP \rrbracket)]$$

Given the denotation of *part* in (28), the truth conditions of (57-a) are as follows, where $\llbracket \text{of } DP \rrbracket$ denotes the set of mereological parts of $\llbracket DP \rrbracket$ under the parthood relation selected by the partitive marker:

$$(59) \quad \begin{aligned} \llbracket \text{Part of } DP \text{ VP} \rrbracket^c = 1 \text{ iff} \\ \exists x [x \in \llbracket \text{of } DP \rrbracket \wedge \llbracket * [C \text{ VP}] \rrbracket^c(x) = 1 \text{ iff} \\ \exists x [x \in \llbracket \text{of } DP \rrbracket \wedge \exists S [x = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq (Cov^c \cap \llbracket VP \rrbracket)]] \\ \text{(i.e., there exists a part } x \text{ of } \llbracket DP \rrbracket \text{ such that } x \text{ is composed of one or more} \\ \text{subparts, each of which is both in } Cov^c \text{ and in } \llbracket VP \rrbracket). \end{aligned}$$

These truth conditions can be simplified to a simple existential statement. Suppose first that some part x of $\llbracket DP \rrbracket$ is both in the cover and in $\llbracket VP \rrbracket$. Then the sentence is true, since x is the sum of the singleton set $\{x\}$, and $\{x\}$ is a non-empty subset of $(Cov^c \cap \llbracket VP \rrbracket)$. Conversely, suppose that some part y of $\llbracket DP \rrbracket$ satisfies the starred

²⁴It is assumed throughout that the relevant alternatives are upward-monotonic. Non-monotonic quantifiers, such as *exactly half* or *between 60% and 80%*, fall outside the scope of the present analysis. This restriction is consistent with different views of how scalar alternatives are generated. On the lexical approach, alternatives are constrained by monotonicity (Horn 1989; Matsumoto 1995), so non-monotonic quantifiers belong to a different scale and do not compete with monotonic ones. On the structural approach (Katzir 2007; Fox and Katzir 2011), complex non-monotonic quantifiers are likewise excluded as alternatives to *part* and *all*, since they are structurally more complex.

predicate. Then there is a non-empty set $S \subseteq (Cov^c \cap \llbracket VP \rrbracket)$ such that $y = \bigoplus S$. Every element of S is a part of y , and since y is itself a part of $\llbracket DP \rrbracket$, transitivity of parthood entails that every element of S is also a part of $\llbracket DP \rrbracket$. Thus, at least one part of $\llbracket DP \rrbracket$ is both in the cover and in $\llbracket VP \rrbracket$. Therefore, the star operator has no effect on the truth conditions of (57-a), which can be stated as follows:

$$(60) \quad \begin{aligned} \llbracket \text{Part of } DP \text{ } VP \rrbracket^c = 1 \text{ iff} \\ \exists x[x \in \llbracket \text{of } DP \rrbracket \wedge x \in Cov^c \wedge \llbracket VP \rrbracket(x)] \\ \text{(i.e., there exists a part } x \text{ of } \llbracket DP \rrbracket \text{ that is both in } Cov^c \text{ and in } \llbracket VP \rrbracket) \end{aligned}$$

Given the denotation of *all* in (29), the truth conditions of (57-b) are as follows:

$$(61) \quad \begin{aligned} \llbracket \text{All of } DP \text{ } VP \rrbracket^c = 1 \text{ iff} \\ \llbracket * [C \text{ } VP] \rrbracket^c (\bigoplus \llbracket \text{of } DP \rrbracket) = 1 \text{ iff} \\ \llbracket * [C \text{ } VP] \rrbracket^c (\llbracket DP \rrbracket) = 1 \text{ iff} \\ \exists S[\llbracket DP \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq (Cov^c \cap \llbracket VP \rrbracket)] \\ \text{(i.e., } \llbracket DP \rrbracket \text{ is the sum of one or more subparts, each of which is both in } Cov^c \\ \text{and in } \llbracket VP \rrbracket) \end{aligned}$$

In light of the truth conditions derived above, contextual equivalence depends both on the properties of the unmodified predicate and on the choice of cover. Covers are generally taken to be contextually supplied, but I do not attempt to determine which cover is salient in each particular context. Instead, I consider the three cover types assumed to be generally available in the present analysis: a power cover, an atomic cover, and a singleton cover. A sentence is predicted to be infelicitous if it violates CQP under every available cover type; if at least one available cover type avoids the violation, an acceptable interpretation is available in principle. In practice, I consider an atomic cover and a power cover as the relevant possibilities; singleton covers will be shown below to invariably produce a violation of CQP in cases of quantification over parts. Other types of covers could in principle prevent contextual equivalence from arising, but I assume that such covers are not ordinarily salient.

Under a power cover, $Cov^c(x)$ holds for all x such that $x \in \llbracket \text{of } DP \rrbracket$. As a result, the truth conditions reduce to the following:

$$(62) \quad \begin{aligned} & \text{Power-cover interpretation} \\ \text{a. } & \llbracket \text{Part of } DP \text{ } VP \rrbracket^c = 1 \text{ iff} \\ & \exists x[x \in \llbracket \text{of } DP \rrbracket \wedge \llbracket VP \rrbracket(x)] \\ & \text{(i.e., there exists a part } x \text{ of } \llbracket DP \rrbracket \text{ that is in } \llbracket VP \rrbracket) \\ \text{b. } & \llbracket \text{All of } DP \text{ } VP \rrbracket^c = 1 \text{ iff} \\ & \exists S[\llbracket DP \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq \llbracket VP \rrbracket] \\ & \text{(i.e., } \llbracket DP \rrbracket \text{ is the sum of one or more subparts, each of which is in } \llbracket VP \rrbracket) \end{aligned}$$

These truth conditions may become contextually equivalent under certain assumptions. For instance, if it is assumed that only the maximal element in the set of parts of $\llbracket DP \rrbracket$ (namely, $\llbracket DP \rrbracket$ itself) can satisfy the predicate, then the existential statement contextually entails the universal one. Such a case was illustrated in Section 1 by

example (12), where only the answer as a whole could be correct or incorrect in the given context. Additional cases are discussed in Subsections 5.1 and 5.3.

Under an atomic cover, the relevant cover consists of the minimal elements of a set of parts of $\llbracket DP \rrbracket$. In some contexts, this set may itself be contextually restricted. In such cases, an element can count as atomic relative to the restricted set even if it has proper parts in the broader mereological domain (Krifka 1989). For example, suppose the context restricts attention to sections of a wall defined along its length. The relevant set may then contain sections or slabs of different lengths, each extending across the full height and width of the wall. The minimal elements of this set are cross-sections at particular points along the wall's length. These cross-sections count as atomic relative to the restricted set, even though they have smaller spatial parts that are not members of that set. Subsection 5.3.1 develops an explicit account of how such contextual restriction can arise.

More generally, I use the following operator, which maps a set P to the subset of its $\sqsubseteq_c$ -minimal elements:

$$(63) \quad \textit{Atomicity relative to a set} \\ {}^{\text{ATOM}_c}P =_{\text{DEF}} \{x \in P : \neg \exists y[y \in P \wedge y \sqsubset_c x]\}$$

For simplicity, in the truth conditions below I apply this operator to the full set of mereological parts of $\llbracket DP \rrbracket$. In this case, relative atomicity coincides with the absolute notion of atomicity defined in (53). Contextually restricted applications of the operator will be introduced where they are needed later in the paper. This yields the following truth conditions:

$$(64) \quad \textit{Atomic-cover interpretation} \\ \begin{aligned} \text{a.} \quad & \llbracket \text{Part of } DP \text{ } VP \rrbracket^c = 1 \text{ iff} \\ & \exists x[x \in {}^{\text{ATOM}_c}(\llbracket \text{of } DP \rrbracket) \wedge \llbracket VP \rrbracket(x)] \\ & \text{(i.e., there exists an atomic part } x \text{ of } \llbracket DP \rrbracket \text{ that is in } \llbracket VP \rrbracket) \\ \text{b.} \quad & \llbracket \text{All of } DP \text{ } VP \rrbracket^c = 1 \text{ iff} \\ & \exists S[\llbracket DP \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq ({}^{\text{ATOM}_c}(\llbracket \text{of } DP \rrbracket) \cap \llbracket VP \rrbracket)] \\ & \text{(i.e., } \llbracket DP \rrbracket \text{ is the sum of one or more atomic parts, each of which is in } \llbracket VP \rrbracket) \end{aligned}$$

Note that under an atomic-cover interpretation, the universal statement is true just in case every atomic element of the entity satisfies the predicate, resulting in a distributive interpretation (Bar-Lev 2025b). As in the case of a power cover, these truth conditions may become contextually equivalent under certain assumptions, as discussed in Subsection 5.3. Further note that in some cases the relevant parthood relation does not induce a set of atoms, in which case an atomic cover is unavailable; this is arguably the case for material parthood (see Subsection 4.2). In other cases, an atomic cover may exist, but the atomic elements cannot satisfy the predicate (see Subsection 5.1). In both situations, however, a power-cover interpretation remains potentially available.

Finally, under a singleton cover, the only member of Cov^c that is part of $\llbracket DP \rrbracket$ is $\llbracket DP \rrbracket$ itself. Ignoring irrelevant elements in Cov^c , this yields the following truth conditions:

- (65) *Singleton-cover interpretation*
- a. $\llbracket \text{Part of } DP \text{ } VP \rrbracket^c = 1$ iff
 $\exists x[x \in \{\llbracket DP \rrbracket\} \wedge \llbracket VP \rrbracket(x)]$
 (i.e., there exists a part x of $\llbracket DP \rrbracket$ that is a member of the singleton containing $\llbracket DP \rrbracket$ and in $\llbracket VP \rrbracket$)
 - b. $\llbracket \text{All of } DP \text{ } VP \rrbracket^c = 1$ iff
 $\exists S[\llbracket DP \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq (\{\llbracket DP \rrbracket\} \cap \llbracket VP \rrbracket)]$
 (i.e., $\llbracket DP \rrbracket$ is the sum of one or more subparts, each of which is both in the singleton containing $\llbracket DP \rrbracket$ and in $\llbracket VP \rrbracket$)

Under this interpretation, existential and universal quantification are necessarily equivalent. The existential statement is true only if $\llbracket DP \rrbracket \in \llbracket VP \rrbracket$, since no proper part can be a member of $\{\llbracket DP \rrbracket\}$. The universal statement imposes no stronger requirement: the only possible choice for S is $\{\llbracket DP \rrbracket\}$, and this set satisfies the condition just in case $\{\llbracket DP \rrbracket\} \subseteq \llbracket VP \rrbracket$ (i.e., $\llbracket DP \rrbracket \in \llbracket VP \rrbracket$). Thus, both statements are true under exactly the same condition. Consequently, these scalar alternatives are always contextually equivalent under a singleton cover, yielding a systematic violation of CQP. For this reason, singleton covers will not be considered further.

The remainder of this section is organized around the three empirical challenges identified in Subsection 3.2. Subsection 5.1 examines existential predicates such as *be inhabited*, whose acceptable interpretation is derived under an atomic cover. It also explains the contrast between quantification-compatible and quantification-resistant existential predicates, as illustrated in (66). Subsection 5.2 turns to nominal predicates based on singular count nouns, such as *be an island* (67), whose acceptable interpretation is instead derived under a power cover. Finally, Subsection 5.3, which constitutes the major part of this section, is concerned with gradable predicates, as in (68). It is argued that, when interpreted as additive, gradable predicates violate CQP under both an atomic and a power cover, and are therefore quantification-resistant.

- (66) a. Most of the island is inhabited.
 b. #Most of the island straddles the equator.
- (67) Mombasa is partly an island.
- (68) a. Most of the trail is narrow.
 b. #Most of the trail is short.

5.1 Existential predicates

Among the three cover types considered here, sentences involving existential predicates can avoid a violation of CQP only under an atomic cover. Consider again the predicate *be inhabited*:

- (69) a. Part of the island is inhabited.
 b. All of the island is inhabited.

Under a power cover, the distinction between existential and universal quantification collapses. Since *be inhabited* is upward monotonic with respect to parthood, if any

part of the island counts as inhabited, then the island as a whole also counts as inhabited. Thus, whenever (69-a) is true, (69-b) is true as well: the latter condition can be satisfied by $S = \{\llbracket \text{the island} \rrbracket\}$. The two alternatives are therefore contextually equivalent under a power cover.

- (70) *Power-cover interpretation*
- a. $\llbracket (69\text{-a}) \rrbracket^c = 1$ iff
 $\exists x[x \in \llbracket \text{of the island} \rrbracket \wedge \llbracket \text{is inhabited} \rrbracket(x)]$
 - b. $\llbracket (69\text{-b}) \rrbracket^c = 1$ iff
 $\exists S[\llbracket \text{the island} \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq \llbracket \text{is inhabited} \rrbracket]$

By contrast, under an atomic cover, the two statements remain distinct. On the spatial-region interpretation considered above, the relevant atomic elements may be point-region parts of the island. Sentence (69-a) is true if some such element counts as inhabited, where inhabitedness depends on the presence of residents within the r -neighborhood of its location (see Subsection 4.2). This does not entail that all the elements of the atomic cover count as inhabited, so (69-b) may still be false.

- (71) *Atomic-cover interpretation*
- a. $\llbracket (69\text{-a}) \rrbracket^c = 1$ iff
 $\exists x[x \in {}^{\text{ATOM}_c}(\llbracket \text{of the island} \rrbracket) \wedge \llbracket \text{is inhabited} \rrbracket(x)]$
 - b. $\llbracket (69\text{-b}) \rrbracket^c = 1$ iff
 $\exists S[\llbracket \text{the island} \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq ({}^{\text{ATOM}_c}(\llbracket \text{of the island} \rrbracket) \cap \llbracket \text{is inhabited} \rrbracket)]$

An intriguing puzzle is why (69-a)–(69-b) are acceptable, whereas (72-a)–(72-b) are not, when evaluated in a context where the island is roughly round and lacks a salient division into parts. Both *be inhabited* and *straddle the equator* are existential predicates: the sum of an inhabited and an uninhabited region is itself inhabited; similarly, the sum of a region that straddles the equator and a region located entirely within one hemisphere lies itself on both sides of the equator.

- (72) a. #Part of the island straddles the equator.
b. #All of the island straddles the equator.

Since both predicates are existential, a power cover yields a violation of CQP, while a singleton cover does so independently of the predicate. An atomic cover is therefore the only one of the three cover types considered here that could distinguish the alternatives. However, unlike *be inhabited*, *straddle the equator* cannot be satisfied by point-region individuals. To straddle the equator, a spatial region must extend on both sides of the equator and therefore have spatial extent in at least one dimension. A point-region individual cannot satisfy this condition, since its location is a singleton: the corresponding point may lie on the equator or on either side of it, but it cannot extend across both sides. Nor is it natural to evaluate the predicate relative to a neighborhood around the point-region individual's location, as with *be inhabited*: *straddle the equator* is a property of the spatial region itself, rather than a property determined by what lies in its vicinity. Consequently, no member of a point-region

atomic cover satisfies the predicate. Both (72-a) and (72-b) are therefore false under this cover and hence contextually equivalent. Since the alternatives are also contextually equivalent under the power and singleton covers, CQP correctly predicts the sentences to be infelicitous.

The predicate *straddle* can nevertheless become quantification-compatible when the context makes available an atomic cover whose members are themselves spatially extended. This is illustrated by (73):

- (73) The ranch stretches along the border between Texas and New Mexico and partly straddles it.

The contrast between (72-a) and (73) can be understood in terms of contextual restriction on the parts included in the cover. In (73), the elongated shape of the ranch makes its longitudinal axis salient and may restrict the cover to parts defined along this axis. The minimal elements of this restricted set are then cross-sections at particular points along the ranch's length, each extending across its width, rather than point-region individuals. These cross-sections count as atomic relative to the restricted set even though they have smaller spatial parts that are not included in it. Some of these cross-sections may straddle the border, while others do not. Sentence (73) can therefore be true if at least one cross-section straddles the border, whereas the corresponding stronger alternative with *entirely* is true only if every such cross-section straddles the border. The two alternatives are therefore not contextually equivalent, and CQP is satisfied. Thus, the contrasts discussed in this subsection further support the view that quantification compatibility is not a fixed lexical property of a predicate, but emerges from the interaction between the lexical semantics of the predicate, the part structure associated with the nominal argument, and the broader context.²⁵

5.2 Singular count nouns as nominal predicates

In contrast to existential predicates, nominal predicates formed from singular count nouns can avoid a violation of CQP under a power cover. Consider (27), repeated here as (74-a), together with the corresponding universal alternative in (74-b).

- (74) a. Mombasa is partly an island.
b. Mombasa is entirely an island.

Under a power-cover interpretation, (74-a) is true if some part of Mombasa qualifies as an island. This is not contextually equivalent to (74-b), which requires Mombasa to be exhaustively composed of parts that are islands. This condition is not satisfied if some portion of Mombasa is part of the mainland, as is in fact the case.²⁶

- (75) *Power-cover interpretation*

²⁵I return in Subsection 5.3 to how contextual restriction on the parts included in a cover can be derived from a salient reference space.

²⁶Note that (74-b) intuitively implies that Mombasa consists of a single island, whereas the truth conditions derived here incorrectly predict it to be true even if Mombasa is composed of multiple islands. I return to this overgeneration problem in Subsection 7.3.

- a. $\llbracket (74\text{-a}) \rrbracket^c = 1$ iff
 $\exists x[x \in \llbracket \text{of Mombasa} \rrbracket \wedge \llbracket \text{is an island} \rrbracket(x)]$
- b. $\llbracket (74\text{-b}) \rrbracket^c = 1$ iff
 $\exists S[\llbracket \text{Mombasa} \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq \llbracket \text{is an island} \rrbracket]$

5.3 Gradable predicates

The distinction between quantification-compatible and quantification-resistant gradable predicates is illustrated by the contrast in (24), repeated here in (76).

- (76)
- a. Most of the Great Wall of China is 6–7 m high.
 - b. Most of the Great Wall of China is 4–5 m wide.
 - c. #Most of the Great Wall of China is 12,000 km long.

This subsection is structured as follows. Subsection 5.3.1 develops a more explicit account of contextual restriction on the parts included in a cover, especially, though not exclusively, for spatial gradable predicates such as those in (76). The account derives this restriction from a contextually salient reference space RS . Subsection 5.3.2 then argues that quantification compatibility depends on whether the associated measure function is interpreted as additive. Finally, Subsection 5.3.3 connects this empirical generalization to CQP (56).

5.3.1 Reference spaces and contextual restriction

The interpretation of spatial gradable predicates involves two spatial parameters: a measurement space MS and a reference space RS (Gawron 2009). The measurement space MS is the space relative to which a property is measured: it may be one-dimensional, as in length, height, or width, two-dimensional, as in area, or three-dimensional, as in volume. The reference space RS determines the spatial dimension or dimensions relative to which the entity is decomposed into relevant parts, thereby constraining its salient part-whole structure for the purposes of interpretation. As will be shown shortly, RS may also be one-, two-, or three-dimensional.

For example, in (77), the predicate *be 5–14 m high* measures vertical extent, so the relevant measurement space MS is the vertical axis. The wall, however, is most naturally decomposed into sections along its longitudinal axis, which therefore serves as the reference space RS . This is plausibly because the wall is an elongated object, and a decomposition into longitudinal sections is more salient than a decomposition into horizontal slices stacked from bottom to top or vertical slices arranged side by side along the width axis. The sentence thus states that the height of the wall at positions along RS (the longitudinal axis) ranges between 5 and 14 meters, as measured in MS (the vertical axis).

- (77) The Great Wall of China is 5–14 m high.

The reference space RS restricts the set of contextually relevant parts from which a cover is constructed. For example, in (77), taking all mereological parts of the wall into consideration would include parts that do not extend from the ground to the top, such as individual bricks or horizontal slices. Such parts, however, are intuitively

irrelevant to the truth conditions. What matters is the maximal vertical extent of the wall at each position along the reference space RS .

Similarly, (78) intuitively means that no relevant part of the wall extending from the ground to the top is less than 2 m high. The sentence may therefore be judged true even though many mereological parts of the wall, such as individual bricks, are less than 2 m high.

(78) No part of the wall is less than 2 m high.

To formalize how the reference space RS restricts the set of relevant parts, I assume that spatial gradable predicates are evaluated relative to the spatial-region interpretation of their arguments, as introduced in Subsection 4.2. Let s_i be the spatial-region individual corresponding to a concrete individual i , and consider its $\sqsubseteq_p$ -parts. Recall that loc maps such individuals to their eigenspaces, modeled as non-empty sets of spatial points. A projection function $proj_{RS}$ maps such sets of points to their projections onto the reference space RS . This induces an equivalence relation $\sim_{RS}$ on spatial-region parts of s_i : two such parts are RS -equivalent just in case their locations have the same projection onto RS .

(79) *Equivalence relation $\sim_{RS}$*
 $a \sim_{RS} b =_{\text{DEF}} proj_{RS}(loc(a)) = proj_{RS}(loc(b))$

Given a set of spatial-region individuals P , some members of P may be proper parts of others while having the same projection onto RS (for instance, a horizontal slice of a vertical slab of the wall). This motivates an operator that selects the RS -maximal members of P :

(80) *RS -maximalization*
 $^{RS\text{-max}}P =_{\text{DEF}} \{x \in P : \neg \exists y[y \in P \wedge x \sim_{RS} y \wedge x \sqsubset_p y]\}$
 (i.e., $^{RS\text{-max}}P$ is the subset of P consisting of those elements that are maximal among the members of P with the same projection onto RS)

For example, let $W = loc(s_w)$ be the wall's eigenspace, and fix an interval $I \subseteq RS$. Many spatial-region parts of the wall have locations that project onto I , including parts corresponding to thin horizontal slices of the wall. The RS -maximal part corresponding to I is the spatial-region individual whose location contains all points in W whose projection onto RS lies in I . A spatial-region individual corresponding to a horizontal slice may have the same projection, but it is not RS -maximal, since there is a larger spatial-region part of the wall with the same projection onto RS .

For the spatial gradable predicates considered here, I take the contextually restricted set of relevant parts to be obtained by applying RS -maximalization to the parts of the spatial-region individual. Thus, RS excludes parts that differ from larger parts only along dimensions that are irrelevant to the reference space. In the wall example, horizontal slices and individual bricks are therefore excluded because they are contained in larger spatial-region parts with the same projection onto the wall's

longitudinal axis. The resulting set consists of the RS -maximal parts that are available for inclusion in the cover. Accordingly, (77) describes the height of the wall across these relevant parts, while (78) requires none of them to be less than 2 m high.

While in the wall case RS is one-dimensional, it may also be two- or three-dimensional, depending on the phenomenon under discussion. Consider (81). Suppose we are interested in how depth varies across the surface of the lake, rather than only along a single direction (as in the wall case). Then it is natural to take the reference space RS to be the two-dimensional surface region of the lake. Let $L = loc(s_l)$ be the lake's eigenspace. Fix a surface region $A \subseteq RS$. There are many spatial-region parts of s_l whose locations project onto A . The RS -maximal part corresponding to A is the spatial-region individual whose location consists of all points in L whose projection lies in A . Intuitively, its location consists of the full vertical columns under A : at each surface location in A , the maximal part extends through the entire depth of the lake at that location.

(81) Most of the lake is shallow.

Now consider (82). Here the relevant reference space RS is naturally three-dimensional. When RS is three-dimensional, projection onto RS preserves the entire location of a spatial-region individual. Since spatial-region individuals are individuated by their locations, no two distinct spatial-region parts can therefore have the same projection onto RS . The RS -maximality condition consequently becomes vacuous: every spatial-region part of the solar system is RS -maximal. Thus, RS does not exclude any spatial-region parts in this case.

(82) Most of the solar system is void.

The role of RS can now be summarized as follows. In quantification over parts of singular individuals, I assume that contextually supplied covers may vary along two dimensions. First, the context determines a reference space RS , and RS -maximalization restricts attention to those parts of $\llbracket of DP \rrbracket$ that are maximal among parts with the same projection onto RS . When RS -maximalization is vacuous, no parts are excluded on this basis. Second, the cover may include all of the remaining parts of $\llbracket of DP \rrbracket$, as in a power cover, or impose a further restriction on them, as in an atomic or singleton cover. In an atomic cover, this further restriction selects the minimal elements of the relevant set of parts; in a singleton cover, only $\llbracket DP \rrbracket$ itself is included. I assume that these two components of cover specification apply in this order, although I do not model their interaction compositionally. Rather, both are treated as aspects of the contextually salient cover introduced by the cover variable.

5.3.2 Empirical generalization

Measure functions can be divided into two classes: additive and non-additive. A measure function is additive if the measure of the mereological sum of two non-overlapping entities is the same as the arithmetic sum of the measures of each entity, as defined in (83). For example, if we consider a swimming pool, volume and mass are additive, whereas temperature and chlorine level are non-additive.

(83) *Additivity*

A measure function μ is additive if and only if:

$$\forall x \forall y [\neg(x \circ y) \rightarrow \mu(x \oplus y) = \mu(x) + \mu(y)]$$

Importantly, while some measure functions are inherently additive or non-additive, others depend on spatial configuration. In particular, spatial dimensions such as length, height, width, and depth may behave additively or non-additively depending on how the relevant objects are arranged. For example, consider two boxes: if they are stacked vertically, the height of their sum equals the sum of their individual heights, whereas if they are placed side by side, this is not the case.

The definition of additivity in (83), however, is too strong for spatial dimensions such as length, height, and width. It requires additivity whenever two entities do not overlap, regardless of the spatial dimension along which they are separated. Thus, height is not additive in the general sense of (83), since the heights of two boxes placed side by side do not sum, even though height behaves additively when the boxes are stacked vertically. To capture this dependence on spatial configuration, I introduce a notion of relativized additivity, defined with respect to a measurement space MS .²⁷

I define relativized additivity as follows. Let x and y range over spatial-region individuals, and let *int* map a set of spatial points to its interior. Two projections are taken to be non-overlapping if their interiors are disjoint. For example, the vertical projections of two boxes stacked one on top of the other count as non-overlapping: although the projections share a boundary, their interiors are disjoint. A measure function associated with a measurement space MS is additive relative to MS if it is additive over spatial-region individuals whose projections onto MS are non-overlapping.

(84) *Relativized additivity*

A measure function μ associated with a measurement space MS is additive relative to MS if and only if:

$$\forall x \forall y [int(proj_{MS}(loc(x))) \cap int(proj_{MS}(loc(y))) = \emptyset \rightarrow \mu(x \oplus y) = \mu(x) + \mu(y)]$$

What matters for quantification compatibility, however, is whether the measure function is additive relative to the reference space RS . A measure function μ is additive relative to RS if, whenever two relevant parts have non-overlapping projections onto RS , the measure of their sum is the sum of their individual measures. If μ is additive relative to MS , then additivity relative to RS follows whenever non-overlap in RS entails non-overlap in MS .

In particular, when $MS = RS$, additivity relative to MS is equivalent to additivity relative to RS . For example, in (85), the reference space RS is the longitudinal axis of the wall, and the measurement space MS coincides with it. Thus, length is additive over relevant parts of the wall whose projections onto RS do not overlap.

(85) The Great Wall of China is 21,196 km long.

²⁷An alternative approach would be to adopt a mereotopological framework, which enriches standard mereology with the topological notion of connectedness. See Grimm (2012a,b) and Wągiel (2021) for linguistic applications.

By contrast, in (86), the measurement space MS is vertical, whereas the reference space RS is longitudinal. Two relevant parts may have non-overlapping projections onto RS while having overlapping projections onto MS . The height of their sum therefore need not equal the sum of their individual heights, and hence height is not additive relative to RS in this context.

(86) The Great Wall of China is 5–14 m high.

Returning to the contrast in (76), I propose the empirical generalization in (87). As discussed in Subsection 5.3.3, this generalization holds in the absence of a contextually salient intermediate cover, since the availability of such a cover may make the sentence acceptable.

(87) *Empirical generalization*
Gradable predicates whose associated measure function is additive relative to the reference space RS are generally incompatible with partitive quantification.

In (76), the reference space RS corresponds to the longitudinal axis of the wall. Relative to this axis, height and width are non-additive, since disjointness of projections onto RS does not entail disjointness of projections onto the vertical or width axes. By contrast, length is additive, because the measurement space MS for length coincides with RS . Consequently, predicates based on height and width are quantification-compatible in this context, while (76-c) is infelicitous.

The generalization also correctly predicts that the quantification compatibility of a gradable predicate does not depend on the monotonicity of the measure phrase. This is illustrated by (25), repeated here as (88). Regardless of the measure phrase, the measure function associated with the predicate remains additive relative to the reference space RS , because the measurement space MS coincides with RS .

(88) #Most of the Great Wall of China is {exactly / more than / less than} 12,000 km long.

Beyond spatial gradable predicates, predicates whose measure functions are inherently additive are predicted to be quantification-resistant regardless of the choice of reference space. Since such functions are additive independently of spatial configuration, they are also additive relative to any relevant RS . For example, (89) is infelicitous because weight is inherently additive: the weight of a whole is the sum of the weights of its non-overlapping parts. By contrast, (90) is acceptable because temperature is inherently non-additive: it does not sum across non-overlapping parts.²⁸

(89) #Most of the box is heavy.

²⁸The distinction between inherently additive and non-additive properties is often described in terms of *extensive* and *intensive* properties, respectively (Krifka 1998; Champollion and Krifka 2016). An extensive property (e.g., mass or volume) is one whose measure for a whole equals the sum of the measures of its parts. By contrast, an intensive property (e.g., temperature or density) does not combine additively across parts, and therefore does not depend on the amount of substance measured. In this paper, however, I adopt the terminology of additivity, in order to accommodate the notion of relativized additivity, which is central to the empirical generalization in (87).

(90) Most of the house is cold.

A further prediction is that spatial gradable predicates, which are not inherently additive or non-additive, should be quantification-compatible in some contexts but not others, depending on their relation to the reference space. This prediction is borne out. In (91), the reference space *RS* is plausibly the scalp area, and hair length is non-additive relative to this space, so the sentence is predicted to be acceptable. This contrasts with the infelicity of (76-c), where *be long* is interpreted as additive relative to the reference space.

(91) Most of Mary's hair is 60 cm long, but she keeps it shorter on her left temple.

Interestingly, it is often difficult to rescue an infelicitous sentence by shifting to a different reference space. For example, (92) is odd because the only salient reference space *RS* is the sidewalk's longitudinal axis. This axis coincides with the measurement space *MS* associated with the predicate, so the predicate is additive relative to *RS*. In principle, one could imagine a different reference space. If *RS* were instead the width axis of the sidewalk, the predicate would no longer be additive relative to *RS*. However, such a decomposition is highly unsalient in this context.

(92) #Most of the sidewalk is 50 m long.

By contrast, a sentence like (93) more readily allows different reference spaces. It may describe a situation in which a long stretch of the sidewalk is completely blocked (i.e., *RS* is the longitudinal axis), and pedestrians are required to cross to the other side of the road, but it can also describe a situation in which a substantial portion of the sidewalk's width is obstructed along the entire street (i.e., *RS* is the width axis), leaving only a narrow path for pedestrians.

(93) Most of the sidewalk is blocked.

Thus, quantification compatibility is not merely a matter of which reference spaces are possible in principle, but of which are contextually salient. In the case of (92), it is unnatural to construe the sidewalk as divided into width-wise sections, each with its own length. Hypothetically, (92) could describe a situation in which a 2-m-wide section of the sidewalk extends for 50 m, whereas a 1-m-wide section extends for only 20 m, so that the sidewalk is 3 m wide for 20 m and then becomes narrower. However, such a partition is not natural in this context, and the sentence therefore remains infelicitous.

5.3.3 Analysis

To account for the generalization in (87), we must show how additivity of the measure function relative to the reference space *RS* can give rise to a violation of CQP (56). Consider the predicate *be 1 km long* in the following context:

- (94) *Context: A straight wall with no salient individuated parts runs for 10 km.*
a. #Part of the wall is (exactly) 1 km long.

- b. #All of the wall is (exactly) 1 km long.

In this case, the reference space RS is the longitudinal axis of the wall, restricting attention to RS -maximal parts. Under a power-cover interpretation, all of these parts are included in the cover, yielding the following truth conditions:

- (95) *Power-cover interpretation with RS -maximalization*
- a. $\llbracket (94\text{-a}) \rrbracket^c = 1$ iff
 $\exists x [x \in {}^{RS\text{-max}}(\llbracket \text{of the wall} \rrbracket) \wedge \llbracket \text{is 1 km long} \rrbracket(x)]$
 - b. $\llbracket (94\text{-b}) \rrbracket^c = 1$ iff
 $\exists S [\llbracket \text{the wall} \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq ({}^{RS\text{-max}}(\llbracket \text{of the wall} \rrbracket) \cap \llbracket \text{is 1 km long} \rrbracket)]$

In words, (94-a) states that there exists an RS -maximal section of the wall that is 1 km long, whereas (94-b) states that the wall can be covered by one or more RS -maximal sections, each of which is 1 km long.

Crucially, these two conditions are contextually equivalent. Since the wall extends continuously along RS and the power cover contains RS -maximal parts of arbitrary extent along this axis, every point of the wall lies in some 1 km-long section. Thus, if there exists one RS -maximal section of the wall that is 1 km long (i.e., if the wall is at least 1 km long), then the set of all such 1 km-long sections forms a cover of the entire wall. As a result, whenever (94-a) is true, (94-b) is true as well.

The same contextual equivalence arises whether the predicate is non-monotonic, as in (94), upward monotonic (96), or downward monotonic (97). For the upward-monotonic predicate in (96), if the wall is at least 1 km long, then every point on RS lies in some RS -maximal section of the wall whose length is at least 1 km. Accordingly, the set of all such sections forms a cover of the wall, so the existential and universal alternatives are contextually equivalent. For the downward-monotonic predicate in (97), every point on RS likewise lies in some sufficiently short RS -maximal section of the wall, namely one whose length is less than 1 km. Here too, the set of all such sections forms a cover of the wall, yielding contextual equivalence between the two alternatives.

- (96) a. #Part of the wall is at least 1 km long.
b. #All of the wall is at least 1 km long.
- (97) a. #Part of the wall is less than 1 km long.
b. #All of the wall is less than 1 km long.

Turning to the atomic-cover interpretation, the first component of the cover remains the same: RS -maximalization restricts attention to the RS -maximal parts of the wall. The atomic-cover component then further restricts this set to its minimal elements. More specifically, an atomic RS -maximal part is an RS -maximal part that has no proper part that is itself RS -maximal. In the case of the wall, this delivers the cross-sections at individual positions along its longitudinal axis. Each such cross-section projects onto a single point on RS while extending maximally in the orthogonal dimensions. This yields the following truth conditions:

(98) *Atomic-cover interpretation with RS-maximalization*

- a. $\llbracket (94\text{-a}) \rrbracket^c = 1$ iff
 $\exists x [x \in {}^{\text{ATOM}_c}({}^{RS\text{-max}}(\llbracket \text{of the wall} \rrbracket)) \wedge \llbracket \text{is 1 km long} \rrbracket(x)]$
- b. $\llbracket (94\text{-b}) \rrbracket^c = 1$ iff
 $\exists S [\llbracket \text{the wall} \rrbracket = \bigoplus S \wedge S \neq \emptyset \wedge S \subseteq ({}^{\text{ATOM}_c}({}^{RS\text{-max}}(\llbracket \text{of the wall} \rrbracket)) \cap \llbracket \text{is 1 km long} \rrbracket)]$

Since each cross-section projects onto a single point on the longitudinal axis, its extent along MS is 0, regardless of the total length of the wall. All atomic elements therefore have the same measure along MS and are indistinguishable with respect to predicates determined solely by that measure. This contrasts with dimensions that are non-additive relative to RS , such as height and width: the same cross-sections have non-zero and potentially varying measures along the vertical and width dimensions.

The fact that all atomic RS -maximal parts have length 0 leads directly to contextual equivalence. If a predicate is true of individuals of length 0 (e.g., *is less than 1 km long*), it is true of every atomic element; if it is false of individuals of length 0 (e.g., *is at least 1 km long*), it is false of every atomic element. Since these atomic elements jointly cover the wall, existential and universal quantification therefore yield the same truth conditions.²⁹

The proposed analysis is supported by the contrast between the infelicity of (94) and the acceptability of the minimal pair in (99). While (99-a) may sound somewhat underinformative, it can be used felicitously in logical reasoning, as in: “Given that the wall is 10 km long, some part of it must be 1 km long.”

- (99) a. Some part of the wall is (exactly) 1 km long.
- b. Every part of the wall is (exactly) 1 km long.

In the absence of a contextually salient partition of the wall, (99-b) is understood as quantifying over parts of arbitrary size. On this reading, the sentence is a contextual contradiction, since the wall must contain parts shorter than 1 km. This does not necessarily lead to a violation of NTA, however, since the scalar alternatives to (99-a) need not be limited to the universal alternative. For example, (100) is neither a contextual contradiction nor a contextual tautology, and it is not contextually equivalent to (99-a). In particular, (100) requires the wall to contain multiple 1 km-long parts and therefore entails that the wall extends for more than 1 km, whereas (99-a) does not. The presence of such a non-trivial alternative is therefore sufficient to satisfy NTA, even if the stronger universal alternative is itself contextually contradictory.

- (100) Many parts of the wall are (exactly) 1 km long.

The contrast supports the proposed analysis because (99-a) and (94-a) have the same truth conditions, suggesting that felicity cannot be determined solely by the informativeness of the assertion. Instead, the crucial difference is that they give rise to different

²⁹Alternatively, additive properties such as length, area, and weight can be analyzed as associated with open scales, which include degrees arbitrarily close to zero but lack a minimum degree (an infimum) (Kennedy and McNally 2005; Kennedy 2007; Gawron 2009). On this view, if a measure function is of type $\langle e, d \rangle$, entities with zero value fall outside its domain. In the wall example, this means that the atomic parts are simply not defined for the property denoted by the predicate.

alternative sets: (99-a) has scalar alternatives with distributive quantifiers, whereas (94-a) has alternatives with non-distributive partitive quantifiers, since it involves quantification over parts of a singular individual. In the present case, unlike (94-a), (99-a) has a non-trivial scalar alternative such as (100), thereby satisfying NTA.

Importantly, additivity relative to *RS* is not by itself sufficient to yield a violation of CQP. Contextual equivalence among scalar alternatives also depends on the cover. Under a power cover, the relevant part structure may be sufficiently rich to contain parts of arbitrary size, so that whenever some part satisfies the predicate, parts that satisfy the predicate can jointly cover the whole. Under an atomic cover, contextual equivalence arises when all minimal elements have the same measure along *MS* and are therefore indistinguishable with respect to the predicate. By contrast, a contextually salient intermediate cover may contain parts with different values along *MS*. In such cases, additivity does not necessarily lead to a violation of CQP.

For example, in (101), the car has a salient partition into components. On the intended interpretation, the sentence states that most of the car, for instance in terms of volume, is lightweight, while a few components, such as the battery, are heavy. Importantly, (101) does not contextually entail the universal alternative. Although the heavy components could in principle be decomposed into smaller, lightweight material parts, those parts are not members of the relevant cover. It therefore does not follow that the car as a whole can be decomposed into lightweight parts in the cover. The existential and universal alternatives are consequently not contextually equivalent, and (101) is acceptable. Similar examples involving object mass nouns and plurals are discussed in the next section.

- (101) Most of this electric car is extremely lightweight, but the battery and a few other components make it heavy.

The empirical generalization in (87) should thus be understood as holding in the absence of a salient cover that introduces relevant variation among parts with respect to the measure associated with the predicate.

6 Extensions

Up to this point, the analysis has focused on quantification over parts of singular individuals. I now extend the discussion to object mass nouns (Subsection 6.1) and to plural collective predication (Subsection 6.2), where the contrast between quantification-compatible and quantification-resistant predicates is known as the *gather/numerous* distinction. I argue that the analysis developed here extends naturally to these domains as well, while also raising several issues.

6.1 Object mass nouns

Object mass nouns such as *furniture* introduce an important qualification to the empirical generalization in (87). Consider (102), which is acceptable on a distributive reading, under which it means that most individual pieces of furniture are each heavy.

- (102) Most of the furniture is heavy.

I assume that the relevant cover in (102) consists of the individual pieces of furniture. This may be analyzed as a contextually salient intermediate cover or, depending on one’s assumptions about the extensions of object mass nouns, as an atomic cover.³⁰ Either way, the proposed analysis predicts the acceptability of (102). The individual pieces of furniture have non-zero and potentially varying values with respect to weight, so they remain potentially distinguishable with respect to the predicate (i.e., some pieces may be heavy while others are not). The relevant scalar alternatives therefore need not be contextually equivalent.

While the distributive reading of sentences like (102) is correctly predicted by the analysis, a potential challenge arises with respect to collective interpretations. Consider (103). In the described context, the predicate *weigh exactly 40 kg* holds of the subplurality consisting of the four chairs, each weighing 10 kg. Thus, there exists a subplurality comprising most of the furniture whose total weight is exactly 40 kg, so the sentence would be predicted to be true on a collective reading. However, the sentence is in fact infelicitous on this interpretation.

- (103) *Context: The furniture consists of a table and four chairs. The table weighs 50 kg, and each chair weighs 10 kg.*
 #Most of the furniture weighs exactly 40 kg.

Importantly, under a power-cover interpretation, no contextual equivalence arises among scalar alternatives. In particular, the universal alternative (i.e., *All of the furniture weighs exactly 40 kg*) is false in the given context: the sum of all the furniture cannot be decomposed into parts each of which weighs exactly 40 kg, since no subplurality containing the table has a total weight of exactly 40 kg. Nevertheless, sentences such as (103) are typically taken to permit only a distributive reading, as in (102).

I suggest that the infelicity of (103) does not stem from a violation of NTA, but rather from the fact that the sentence addresses an implausible question under discussion (QUD), namely: “Does some, most, or all of the furniture weigh exactly 40 kg?” Support for this view comes from (104). This sentence is felicitous under both a distributive and a collective interpretation. It may be true if most individual pieces each cost less than \$1,000 (“atomic-cover” interpretation), or if there is a subplurality comprising most of the furniture whose total cost is less than \$1,000 (power-cover interpretation).

- (104) Most of the furniture in Bill’s apartment cost him less than \$1,000.
 a. Distributive reading (available): Most individual pieces of furniture each cost less than \$1,000.
 b. Collective reading (available): There is a subplurality comprising most of the furniture such that the total cost of that subplurality is less than \$1,000.

³⁰In (102), *most* is understood in terms of cardinality, even though object mass nouns do not combine with count quantifiers and numerals, as in *#Three of the furniture are heavy* (Bale and Barner 2009; Rothstein 2017, among others). I leave open how this restriction should be reconciled with the present cover-based analysis.

This availability of a collective reading is consistent with the present analysis. Specifically, it does not follow from the collective reading of (104) that the universal alternative (i.e., *All of the furniture in Bill's apartment cost him less than \$1,000*) is true as well. For instance, it may be that some piece of furniture is particularly expensive, so that the sum of all the furniture cannot be decomposed into parts each of which costs less than \$1,000. Crucially, the collective reading is permitted despite the additive nature of the predicate, because no contextual equivalence arises, and NTA is therefore not violated.

The felicity of the collective interpretation in (104) contrasts with the minimal pair in (105). Unlike (104), this sentence is only acceptable under a distributive reading. The reason is that, given additive cost, *cost more than \$1,000* is upward monotonic with respect to the relevant parthood relation. Once some subplurality costs more than \$1,000, every larger subplurality containing it, including the maximal plurality, does as well. The distinction between the existential and universal alternatives therefore collapses under the power cover, violating NTA.

- (105) Most of the furniture in Bill's apartment cost him more than \$1,000.
- a. Distributive reading (available): Most individual pieces of furniture each cost more than \$1,000.
 - b. Collective reading (unavailable): There is a subplurality comprising most of the furniture such that the total cost of that subplurality is more than \$1,000.

The contrast between (104) and (105) shows that the monotonicity of the measure phrase can matter when the relevant cover contains parts with different values with respect to the associated measure. While upward-monotonic predicates, such as *cost more than \$1,000* (105), lead to a violation of NTA, the same does not apply to downward-monotonic predicates, as in (104). This contrasts with the wall example discussed in Subsection 5.3, where contextual equivalence arose either because a power cover contained parts of arbitrary extent or because all elements of the atomic cover had the same value along the relevant measurement dimension.

Non-monotonic gradable predicates are likewise predicted not to violate NTA. For example, (106) seems to permit a collective reading.

- (106) Most of the furniture in Bill's apartment cost him {exactly / about} \$1,000.
- a. Distributive reading (available): Most individual pieces of furniture each cost {exactly / about} \$1,000.
 - b. Collective reading (available): There is a subplurality comprising most of the furniture such that the total cost of that subplurality is {exactly / about} \$1,000.

The acceptability of (104) and (106) contrasts with the infelicity of (103), even though the latter likewise does not violate NTA. The difference, I propose, lies in the plausibility of the QUD. In (104), the relevant QUD ("Did some, most, or all of the furniture in Bill's apartment cost him less than \$1,000?") is a natural and contextually reasonable question, supporting the felicity of the sentence. In contrast, the QUD in (103)

is markedly less plausible as a conversational goal. In a suitable context, however, the sentence may improve. Suppose that a moving company charges a fixed amount for a load weighing up to 40 kg, but imposes a substantial surcharge if the load exceeds this limit. A customer therefore wants to include as much of the furniture as possible in a single load without exceeding 40 kg. In this context, it becomes relevant to determine how much of the furniture can be included in a single load while approaching the 40 kg limit as closely as possible, making exact total weights around 40 kg conversationally salient.

The availability of a collective reading in these cases thus appears to be constrained by two factors: (i) compliance with NTA and (ii) the plausibility of the relevant QUD. These should not necessarily be viewed as entirely independent constraints. NTA itself can be understood as restricting the set of contextually licit QUDs, since a QUD whose relevant answers are contextually equivalent fails to serve a communicative purpose (Bar-Lev and Fox 2023).

6.2 Collective predicates

The empirical generalization in (87) also largely extends to plural collective gradable predicates. In particular, collective predicates based on the additive cardinality measure, such as *be numerous* and *be few in number*, are quantification-resistant. This is illustrated by predicates like *be numerous* (107) and *be few in number* (108). The reason is that, if a subplurality of a given cardinality satisfies the predicate, then for every individual in the maximal plurality there is a subplurality of the same cardinality that contains that individual. All of these subpluralities satisfy the predicate, and together they cover the maximal plurality. As a result, under a power-cover interpretation, whenever an existential statement such as (108-a) is true, the corresponding universal alternative (108-b) is true as well. This yields contextual equivalence between existential and universal quantification, and the infelicity of (107)–(108) is therefore correctly predicted by the analysis.

- (107) a. #Some of the kids were numerous.
 b. #All of the kids were numerous.
- (108) a. #Some of the kids were few in number.
 b. #All of the kids were few in number.

A different pattern emerges with collective predicates based on non-additive measures. In principle, these should be quantification-compatible, yet this prediction is not fully borne out. For example, although density is non-additive, the sentence in (109) is somewhat degraded. This contrasts with (110), where the quantifier ranges over parts of a singular individual, and the sentence is fully acceptable.

- (109) #Most of the trees in the grove are dense.
- (110) Most of the grove is dense(ly wooded).

I suggest that the infelicity of (109) might stem from a mismatch between the domain of quantification and the requirements of the predicate. Specifically, the quantifier ranges over pluralities of trees, while *be dense* is not naturally interpreted as a property

of such pluralities alone: density must be evaluated relative to some spatial extent, such as an area occupied by the trees. In contrast, (110) involves quantification over spatial-region parts corresponding to areas of the grove, making density straightforwardly interpretable.

This explanation is further supported by the fact that (109) becomes acceptable under a slight modification in which the predicate directly characterizes the spatial arrangement of the trees. In this case, the sentence becomes natural, as shown in (111).

- (111) Most of the trees in the grove were too densely planted.

Other non-additive gradable predicates that characterize the spatial configuration of plural individuals are likewise quantification-compatible. These include complex predicates such as (112)–(113) and lexical predicates such as *be crowded* in (114) (where *crowded* means that the teeth do not have enough space to fit in the jaw). In all of these cases, the relevant measure is non-additive: its value for a plurality is not determined by summing the values of non-overlapping subpluralities. These predicates are therefore predicted to be compatible with quantification over parts.

- (112) Most of the glasses are tightly packed inside the box.

- (113) Most of the buildings on this street are closely spaced.

- (114) Most of Bill's teeth are crowded.

Beyond gradable predicates, a central question regarding the *gather/numerous* distinction concerns contrasts such as the one in (115). Under the assumption that the plural predicate is modified by a star operator (Link 1983), the sentence in (115-a) with a definite plural subject is interpreted collectively: it states that each judge is part of a plurality of judges who convicted the defendant. This raises the question of why the same meaning cannot be expressed with an overt universal quantifier, as in (115-b). Notably, (115-b) seems to force a coerced interpretation of *convict* as 'vote to convict', yielding a distributive reading over individual judges rather than the intended collective one.

- (115) *Context: In a certain judicial system, exactly five judges sit in every case. The verdict is formally returned by the panel as a whole and is decided by a simple majority.*

- a. The judges convicted the defendant.

- b. #All the judges convicted the defendant.

Under the proposed analysis, this contrast follows from differences in the relevant alternative sets. Specifically, the sentence in (115-b) is infelicitous because it violates NTA. Given the context, only the maximal plurality of judges can return a verdict. As a result, existential and universal quantification become contextually equivalent: if there exists a plurality of judges who convicted the defendant, it must be the maximal one.³¹

³¹This equivalence holds regardless of whether one assumes a power cover or a singleton cover.

By contrast, the definite plural in (115-a) evokes a different alternative set, which includes the corresponding bare plural *Judges convicted the defendant*. The definite description is interpreted relative to a contextually restricted domain containing the judges who constitute the panel and refers to the maximal plurality in that domain. If the same domain restriction is assumed for the bare plural, the definite and bare-plural alternatives are contextually equivalent. Nevertheless, the definite is favored over the bare plural by Maximize Presupposition (Heim 1991), since the presuppositional requirements of the definite are satisfied in the context. Thus, the contextual equivalence between these alternatives does not give rise to the NTA effect found with scalar alternatives.³²

The proposed analysis may also offer insight into a class of accomplishment predicates such as *build the raft*, which tend to be quantification-compatible despite lacking stratified reference (Taub 1989). Kuhn (2020, 243) reports, based on previous unpublished results, that sentences like (116) receive intermediate naturalness ratings, falling between prototypical *gather*-type and *numerous*-type predicates. Kuhn classifies these predicates as members of the *numerous* class, but suggests that a particularly accessible rescue mechanism makes accomplishment predicates more acceptable in quantificational contexts.

(116) All of the kids built the raft.

By contrast, under the present analysis, accomplishment predicates of this type are expected to be quantification-compatible. The reason is that the truth of an existential statement does not force the corresponding universal statement to be true. If some subplurality of kids built the raft, it does not follow that the maximal plurality of kids can be decomposed into subpluralities each of which built the raft. Accordingly, no contextual equivalence is predicted between existential and universal quantification.

However, the picture is apparently complicated by the availability of a team-credit (Dowty 1987) or collective-action (Landman 2000) interpretation. Under such a reading, the sentence in (117) may be judged true even in a situation where only a few of the kids actually did the work, while the others merely stood by. In that case, the maximal plurality may count as having built the raft as a team. This kind of interpretation can weaken the contrast between existential and universal quantification, which may explain why (116) is only marginally acceptable rather than fully natural.

(117) The kids built the raft.

Finally, another case in which the proposed analysis yields a different prediction from previous accounts concerns sentences like (118). Nominal predicates based on singular count nouns, such as *be a good team*, are generally considered prime examples of *numerous*-type predicates (Winter 2002a). Crucially, (118) does not violate NTA: it is possible for a subplurality of the students to constitute a good team while the maximal plurality of students cannot be decomposed into subpluralities each of which constitutes a good team. Why, then, is this sentence nevertheless infelicitous, unlike

³²See the appendix for a comparison with the analysis proposed by Winter (2002a) regarding the contrast between definite plurals and overt universal quantifiers.

its counterpart with *be an island* in (27), which likewise involves a singular count noun functioning as a nominal predicate?

(118) #All the students are a good team.

Once again, I suggest that such sentences are not inherently unacceptable but become infelicitous because they address an implausible QUD, much like (103). In the case of (118), the underlying QUD is “Are some, most, or all of the students a good team?”, which might be perceived as somewhat unnatural. On this view, the felicity of such sentences depends on the availability of a plausible QUD. This seems to be the case in (119), which is acceptable, presumably because the team of mountaineers is already individuated from the rest of the hotel guests, making it natural to ask what proportion of the guests constitutes that group.

(119) Most of the guests at this hotel are a team of mountaineers who are setting out before dawn tomorrow.

7 Remaining issues

7.1 Overt quantification vs. definite subjects

So far, the discussion has focused on sentences with overt partitive quantifiers. A natural question is how this analysis relates to sentences without overt quantifiers, such as those with definite subjects. These sentences are known to differ from their overtly quantified counterparts in at least two important respects: they typically exhibit homogeneity and permit non-maximal interpretations.

Homogeneity refers to the fact that affirmative and negative sentences with definite subjects often fail to yield complementary truth conditions. For example, since Ecuador straddles the equator, neither (120-a) nor (120-b) is straightforwardly true:

- (120) a. Ecuador is in the Southern Hemisphere.
b. Ecuador is not in the Southern Hemisphere.

Non-maximality is illustrated by examples such as (121-a). The sentence may be judged true even though not all of Nepal is mountainous, as Nepal also contains lowland areas. By contrast, the overtly quantified sentence in (121-b) lacks this non-maximal reading and is judged false.

- (121) a. Nepal is mountainous.
b. All of Nepal is mountainous.

Overt universal quantification with *all* has been argued to remove homogeneity (Löbner 2000), and thereby also remove non-maximality (Brisson 2003; Križ 2016). This effect does not straightforwardly follow from the present analysis. Given the bivalent star operator assumed here, (121-a) and (121-b) receive the same truth conditions: both are true just in case Nepal can be decomposed into one or more parts each of which is mountainous.

One possible way forward would be to adopt, following Bar-Lev (2025b), a trivalent star operator to derive the homogeneity and non-maximality of (121-a), rather than the bivalent star operator assumed in the current analysis. The remaining challenge would be to explain why the falsity and undefinedness conditions introduced by the trivalent star operator collapse when the predicate appears in the nuclear scope of an overt quantifier such as *all*, as in (121-b), resulting in the homogeneity-removal effect. I leave the integration of the present proposal with such an account to future research.

7.2 The relative acceptability of *all*

As pointed out by an anonymous reviewer, *all* is sometimes more acceptable with quantification-resistant predicates than other quantifiers such as *most*, *part*, and *some*. For example, (122-a) seems relatively acceptable, despite involving the additive predicate *cost \$50* (see Subsection 5.3). By contrast, (122-b) is unacceptable.

- (122) *Context: Bob tells a friend how cheap gas was in the country where he spent a two-week vacation.*
- a. All the gas I used during the vacation cost me \$50!
 - b. #Most of the gas I used during the vacation cost me \$40!

A similar pattern is observed in (123). The predicate *be enough* is typically quantification-resistant with respect to its first argument. Nevertheless, this predicate can sometimes combine with *all*, as in (123-a), attested in the Corpus of Contemporary American English (COCA; Davies 2008). By contrast, the corresponding sentences with *most* and *some* are degraded, as shown in (123-b)–(123-c). Crucially, (123-a) receives a non-distributive interpretation, according to which the running around is sufficient in aggregate to give a man a heart attack. In contrast, to the extent that (123-b) and (123-c) are acceptable, they are interpreted as distributing over types of running around, and thus do not permit a collective or cumulative reading.

- (123) a. All this running around is enough to give a man a heart attack.
[COCA: Fiction, 1993]
- b. #Most of this running around is enough to give a man a heart attack.
 - c. #Some of this running around is enough to give a man a heart attack.

Following Champollion (2017, 244), I suggest that these examples may involve a non-quantificational use of *all*. In such cases, *all* appears to function as an intensifying modifier rather than as a genuine quantifier. For instance, in (123-a), *all* conveys that there was a lot of running around.³³ Similarly, in (122-a), the acceptability of *all* appears to depend on a comparable intensificational effect: the sentence conveys that a relatively large amount of gas was consumed. Indeed, the sentence becomes somewhat odd in contexts where this “large quantity” implication is not satisfied—for example, if the speaker is instead complaining that gas was expensive and that even the small amount he used cost him \$50.

³³In this usage, *all* tends not to co-occur with *of*, although this is not a strict restriction, as illustrated by the following example:

(i) All of this running about has given me quite a thirst. [COCA: Fiction, 1999]

A key diagnostic for this use of *all* is that it is restricted to prenominal position. This is illustrated by the contrast in (124). While (124-a) conveys that the person smoked a large number of cigarettes, the version with adverbial *all* in (124-b) strongly favors an implausible distributive interpretation, in which each cigarette individually wrecked the person’s voice.

- (124) a. All those cigarettes he smoked wrecked his voice.
 b. #Those cigarettes he smoked all wrecked his voice.

This pattern contrasts with quantification-compatible predicates, which may co-occur with either determiner or adverbial *all*, as shown in (125) (see Section 2).

- (125) a. All the kids gathered outside.
 b. The kids all gathered outside.

In light of these data, it is possible that non-quantificational uses of *all*, which are beyond the scope of this paper, interfere with the evaluation of examples involving prenominal *all*. To determine whether a predicate is genuinely quantification-compatible in a given context, it is therefore preferable to consider sentences with other determiners, such as *most* and *part*, or with adverbial modifiers such as *all* and *entirely*.

A different possibility is that the relative acceptability of *all* reflects a more general asymmetry between logically stronger and weaker alternatives that are contextually equivalent. In particular, Magri (2009) argues that the logically weaker expression triggers an obligatory scalar implicature in such situations, giving rise to a contextual contradiction. The alternative with *all*, by contrast, competes with the corresponding alternative with *the*, which carries a homogeneity presupposition. Asserting the alternative with *all* therefore gives rise to the strengthened presupposition that the homogeneity presupposition of the corresponding alternative with *the* is false. When this strengthened presupposition contradicts the common ground, the alternative with *all* is odd. In the cases considered here, the contexts in which the existential and universal alternatives are contextually equivalent are precisely those in which the common ground entails the homogeneity presupposition associated with the corresponding definite plural. Thus, the weaker and stronger alternatives are predicted to be infelicitous in the same contexts (provided that the relevant homogeneous alternative is available). However, because their infelicity is attributed to somewhat different mechanisms, this account leaves open the possibility that they differ in their degree of acceptability.

Spector (2014), by contrast, proposes a symmetric account. On Spector’s version of NTA (7), adopted in this paper, logically weaker and stronger alternatives are both predicted to be infelicitous when they are contextually equivalent. The relative acceptability of *all* in the examples above therefore cannot be derived from NTA alone. However, this contrast is not sufficiently robust to provide strong evidence for choosing between the symmetric and asymmetric approaches.

7.3 Overgeneration of cumulative (intermediate) readings

An anonymous reviewer points out that extending phrasal cumulativity to singular predication (see Subsection 4.2) generates interpretations that do not always appear to be available. Consider (126). Under a power-cover interpretation, phrasal cumulativity predicts that the sentence is true if Vanuatu can be decomposed into parts, each of which is an island. Intuitively, however, the sentence means that Vanuatu as a whole constitutes a single island, rather than being composed of multiple islands. Given that Vanuatu in fact consists of many islands, the sentence is false or undefined.

(126) All of Vanuatu is an island.

The same problem arises with more complex predicates containing singular indefinites or numerals. Consider (127). Under a power-cover interpretation, phrasal cumulativity predicts that the sentence is true if the wall can be decomposed into parts, each of which was built in two years. However, this condition can be satisfied even if building the wall as a whole took considerably longer than two years. The predicted truth conditions are therefore too weak: the sentence is naturally understood as asserting that the construction of the entire wall was completed within two years.

(127) All of the wall was built in two years.

A similar issue arises with definite subjects. As shown in (128), the plural sentence readily allows a distributive interpretation: (128-a) can describe a situation in which different stories are set on different desert islands (e.g., *Robinson Crusoe* and *Lord of the Flies*). By contrast, (128-b) is naturally understood as asserting that the story takes place on a single island. It does not appear to allow the corresponding part-based interpretation on which different parts of the story are set on different islands. Thus, extending phrasal cumulativity from plural to singular predication predicts a reading for (128-b) that does not seem to be available.

- (128) a. These stories are set on a desert island.
b. This story is set on a desert island.

The overgeneration problem just illustrated is part of a broader problem that has long been discussed in the literature on distributivity and cumulativity. In plural predication, intermediate readings, which fall between collective and atomic distributive interpretations, are available in some cases but not others. For instance, suppose that six students divide into two groups of three, and each group independently solves the same problem. In this situation, (129) can naturally be judged true, even though the students did not all solve the problem together and no individual student solved it alone. By contrast, (130) is not ordinarily judged true in a situation in which the students divide into two groups and each group builds a different raft.

(129) The students solved the problem.

(130) The students built a raft.

The theoretical challenge is therefore to allow intermediate readings in cases like (129) without overgenerating them in cases like (130). Crucially, if the star operator applies obligatorily at the VP level, as assumed in the current paper, algebraic closure generates sums of individuals already in the extension of the VP, predicting intermediate readings that are often unavailable (Winter 2002b; Kratzer 2008; Champollion 2016, 2017, among others).³⁴

A closely related problem for pervasive application of the star operator arises with *for*-adverbials. For example, (131) cannot normally describe a situation in which the speaker petted different rabbits successively over a total period of two hours; instead, the same rabbit is understood to be involved throughout the relevant interval (Zucchi and White 2001; van Geenhoven 2004; Kratzer 2008; Champollion 2016, among others).

(131) I petted a rabbit for two hours. (Kratzer 2008)

One approach to this overgeneration problem is to restrict the distribution of the star operator. Kratzer (2008) argues that phrasal cumulativity is restricted to predicates that are sisters of plural DPs. Thus, the star operator does not pluralize the predicate *petted a rabbit* in (131). In plural predication, by contrast, Kratzer assumes phrasal cumulativity but argues that the availability of intermediate readings is sensitive to event individuation: the relevant subevents must be sufficiently well individuated in the context. Champollion (2016) adopts a still more restrictive position, assuming lexical cumulativity of verbs while rejecting unrestricted phrasal cumulativity of VPs. Phrasal distributive readings are instead derived by covert distributivity operators, with non-atomic distributivity requiring a contextually salient cover. Thus, Champollion rules out the unavailable intermediate readings of (130) and (131), while deriving an intermediate reading of (129) from lexical cumulativity.

Despite their differences, under both proposals, the star operator does not apply to the predicates in (126) and (127). Consequently, (126) would require Vanuatu as a whole to be in the extension of *is an island*, while (127) would require the wall as a whole to be in the extension of *was built in two years*. The unwanted part-based intermediate readings would therefore not arise.

Such a restriction is not straightforwardly compatible with the present proposal, however, because the analysis of CQP (6) relies on obligatory application of the star operator in singular predication. In particular, in cases where contextual equivalence depends on phrasal cumulativity, if the star operator did not apply, the relevant scalar alternatives would fail to become equivalent. Consider again (132) under a power-cover interpretation. Without the star operator, (132-a) still states that there is a 1 km-long part of the wall, whereas (132-b) states that the wall as a whole is 1 km long. The two alternatives are therefore not contextually equivalent, and their infelicity would no longer follow from the proposed analysis.

(132) a. #Part of the wall is (exactly) 1 km long.

³⁴A parallel issue arises in an approach that derives intermediate readings through existential quantification over minimal covers (Gillon 1987), since this analysis also predicts such readings in contexts where they are unavailable (Lasnik 1989). For refinements of the cover-based approach that address this overgeneration problem, see Schwarzschild (1996) and Champollion (2016, 2017), among others.

b. #All of the wall is (exactly) 1 km long.

In principle, the star operator could instead apply only to a subset of predicates in singular predication, remaining available in cases like (132) but not in cases like (126) and (127). It is unclear, however, what independent constraint on its distribution would yield this distinction. I therefore do not pursue this possibility here.

A different approach is suggested by Bar-Lev (2025a), who proposes that the absence of certain intermediate readings may follow from scalar implicatures rather than being taken as evidence against unrestricted phrasal cumulativity. In (130), phrasal cumulativity predicts the sentence to be true in an intermediate situation in which two different subgroups of the six students built two different rafts. Crucially, however, the stronger alternative *The students built two rafts* is also true in this situation. Negating this alternative as a scalar implicature yields the inference that the six students built only one raft between them, thereby excluding the intermediate situation and deriving the intuitive interpretation. Thus, on this account, the overgeneration problem does not require restricting the application of the star operator.

A similar account might extend to (126), if the predicative indefinite *an island* gives rise to the corresponding numeral alternative *two islands*. Suppose that Vanuatu is composed of several islands. Under a power-cover interpretation, (126) is true because Vanuatu can be decomposed into parts each of which is an island. But in such a situation, the stronger alternative with *two islands* is also true: Vanuatu can likewise be decomposed into larger parts, each consisting of two islands. Exhaustification therefore yields the inference that Vanuatu can be decomposed into parts each of which is an island, but not into parts each of which consists of two islands. Given the unrestricted power cover assumed here, this enriched meaning can only be satisfied if Vanuatu consists of a single island, thereby excluding the intermediate situation.

A parallel line of reasoning may apply to (127), although less straightforwardly. The sentence has stronger scalar alternatives such as *All of the wall was built in one year*. Under a power-cover interpretation, (127) is true if the wall can be decomposed into parts each built within two years, whereas the stronger alternative is true if it can be decomposed into parts each built within one year. If the relevant temporal interval is allowed to vary freely from one part to another, then for the prejacent to be true while the stronger alternative is false, there must be some part of the wall that cannot itself be further decomposed into parts each built within one year. Given a sufficiently fine-grained part structure, such a situation seems implausible. By contrast, if the relevant temporal interval is held fixed across the parts, the prejacent can naturally be true while the stronger alternative is false: the wall may have been built during a particular two-year interval without having been built during any corresponding one-year interval. Exhaustification may therefore favor an interpretation on which the temporal interval is held constant across parts, yielding the intended reading.

Importantly, however, cumulative readings in singular predication are not categorically unavailable. In some cases, an indefinite can covary across parts quite naturally. For instance, (133-a) can be understood as saying that, throughout most of the sailor's adult life, he was on some boat or other, not necessarily the same boat throughout. Similarly, (133-b) can describe a situation in which different stretches of the trail run

alongside different roads. These cases show that covariation across parts is available in singular predication in principle and are therefore compatible with phrasal cumulativity, although the availability of such interpretations is constrained by contextual factors.

- (133) a. The sailor spent most of his adult life on a boat.
b. Most of the trail runs alongside a road.

One possible factor is the contextual relevance of scalar alternatives. In (133-a), for example, the number of boats may not matter when the question is whether the sailor spent most of his adult life on land or at sea. Similarly, in (133-b), the number of roads may be immaterial when the question is whether the trail runs through the wilderness or alongside developed areas. The stronger alternatives that would otherwise give rise to scalar inferences when excluded may therefore not be deemed contextually relevant.

Similar readings are also available with *for*-adverbials involving stative predicates. In such cases, the value of an indefinite may covary over the relevant temporal interval. For instance, (134) can naturally describe situations in which different boats, different Republican senators, or different suits are involved at different times.

- (134) a. The sailor lived on a boat for 20 years.
b. This Senate seat has been held by a Republican for 80 years.
c. Bill wore a suit to work for 20 years.

With accomplishment and achievement predicates, by contrast, comparable covarying readings are known to be more restricted. As Champollion (2016, 4) observes, (135-a) permits a non-atomic distributive reading in which the indefinite covaries when the relevant period is understood as partitioned into salient subintervals, such as days. By contrast, (135-b) does not normally allow such a reading in the absence of a suitable partition.

- (135) a. The patient took two pills for a month and then went back to one pill.
(Champollion 2016)
b. #John found a flea for ten minutes. (Zucchi and White 2001)

In sum, a promising direction is to investigate whether the overgeneration of phrasal cumulativity in singular predication can be explained by scalar implicatures, as proposed by Bar-Lev (2025a) for plural predication. This possibility is compatible with the observation that cumulative readings with covarying indefinites are available in at least some cases, both with quantification over parts (133) and with *for*-adverbials involving stative predicates (134). These cases suggest that phrasal cumulativity is not categorically excluded in singular predication, although the factors governing its availability remain to be determined.³⁵

³⁵ A different approach to the availability of covarying readings of indefinites has been developed primarily in the literature on *for*-adverbials. Champollion et al. (2017) argues that indefinite noun phrases differ in whether their value may vary across subintervals: flexible indefinites permit different values in different subintervals, whereas rigid indefinites require the same individual to remain fixed throughout. Chierchia (2022, 2023) develops a related idea in terms of a “same participant” constraint on durative modification, requiring the relevant participants to remain constant across the cells of a cover. Whether restrictions of this kind can be extended to quantification over parts remains an open question.

8 Conclusion

This paper has investigated the contrast between predicates that permit quantification over parts and those that resist it. I have argued that quantification compatibility is not a fixed lexical property of predicates. Rather, it depends on the interaction between the predicate and the part structure made salient by the nominal argument and the context. The proposed analysis derives the constraint on quantification over parts (6) as an instance of NTA (7): a quantificational sentence is infelicitous when existential, universal, and intermediate alternatives become contextually equivalent.

The primary focus of the analysis was quantification over parts of singular individuals, covering existential predicates (Subsection 5.1), singular count nouns as nominal predicates (Subsection 5.2), and gradable predicates (Subsection 5.3). The analysis was then extended to object mass nouns (Subsection 6.1) and collective predication (Subsection 6.2). In these latter two domains, some sentences that satisfy NTA were nevertheless found to be infelicitous. I suggested that this infelicity may reflect the implausibility of the relevant QUD, since analogous sentences are acceptable when an appropriate QUD is contextually supported.

Several questions remain unresolved (Section 7). Furthermore, it remains to be seen whether the pragmatic approach proposed here can be extended beyond quantification over parts to other phenomena involving the *gather/numerous* distinction, such as restrictive modification (Pinton 2022; Guerrini and Pinton 2024), or to phenomena that have been analyzed in terms of stratified reference, such as *for*-adverbials (Champollion 2010, 2017). More generally, the present proposal raises the possibility that at least some phenomena previously attributed to lexical constraints on quantificational constructions may instead emerge from contextual equivalence among scalar alternatives.

Appendix: Comparison with Winter (2002a)

While most studies on the *gather/numerous* distinction focus on the mereological properties that distinguish these two classes of predicates, Winter (2002a) is primarily concerned with explaining their compatibility or incompatibility with quantification. He proposes a lexically encoded distinction between atom predicates and set predicates, which cuts across the traditional distributive/collective taxonomy. When not pluralized, atom predicates, such as *smile*, *be a student*, and *convict the defendant*, denote sets of atoms, which may include group atoms. Set predicates, such as *meet*, *gather*, and *be similar*, denote sets of sets of individuals. In this classification, *gather*-type collective predicates are set predicates, while *numerous*-type predicates are atom predicates. The contrast between atom and set predicates emerges when these predicates are combined with non-distributive quantifiers like *all*, as illustrated in (136).

(136) *Context: In a certain judicial system, exactly five judges sit in every case. The verdict is formally returned by the panel as a whole and is decided by a simple majority.*

a. All the judges met in the deliberation room.

- b. #All the judges convicted the defendant.

Suppose there are five judges—Alice, Ben, Carla, David, and Emma—who met in the deliberation room and convicted the defendant. In Winter’s analysis, (136-a) requires every judge to participate in a meeting of judges. Since all five judges belong to the set $\{a, b, c, d, e\}$ and this set is in the predicate’s extension, the sentence is true. By contrast, (136-b) is unacceptable because *convict the defendant* is an atom predicate. Its atomic extension may contain a group atom corresponding to the panel of judges. Crucially, however, this group atom is distinct from the set $\{a, b, c, d, e\}$ of individual judges. Pluralization of an atom predicate does not make the latter set satisfy the predicate; rather, it derives pluralities whose atomic members individually satisfy the predicate. Consequently, the interpretation of (136-b) is equivalent in the relevant respect to that of *Every judge convicted the defendant*.³⁶ This interpretation cannot be satisfied in the context described, in which only the panel as a whole can formally convict the defendant.

As discussed in Section 6, a natural question is why (137) is acceptable, while (136-b) is not. On Winter’s account, the set $\{a, b, c, d, e\}$ and the corresponding group atom (i.e., the panel of judges) are distinct semantic objects. A non-quantificational plural NP such as *the judges* can undergo a set-to-atom mapping and thereby refer to the corresponding group atom, which may be included in the extension of *convict the defendant*. A quantificational NP such as *all the judges*, by contrast, cannot be construed as referring to a group atom. Thus, the infelicity of (136-b) arises because the quantificational subject cannot provide the kind of argument that the predicate demands.

- (137) The judges convicted the defendant.

Under the proposed analysis, on the other hand, the infelicity of (136-b) arises from contextual equivalence among scalar alternatives, since only the maximal plurality can be in the extension of the predicate. In addition, the contrast between (136-b) and (137) is accounted for by the fact that the two sentences evoke different alternative sets, as discussed above in connection with (115). In this way, the present analysis captures both the contrast in (136) and the contrast between (136-b) and (137), without positing Winter’s lexical distinction between atom and set predicates or a special set-to-atom mapping restricted to non-quantificational noun phrases.

In what follows, I demonstrate that the contextual equivalence between the scalar alternatives in (138) also follows under Winter’s analysis of plural quantification, without assuming his lexical distinction between atom and set predicates.

- (138) a. #Some of the judges convicted the defendant.
b. #All the judges convicted the defendant.

Building on Scha (1981) and van der Does (1993), Winter assumes that plural quantifiers like *all* are synonymous with singular quantifiers like *every* (cf. Link 1983).

³⁶This interpretation is analogous to that of *All the judges smiled*, which also involves a pluralized atom predicate. On Winter’s analysis, the sentence is true just in case every judge belongs to a plurality of judges such that each of its members smiled.

On his set-theoretic analysis of plurals, this leads to a type mismatch between the extension of the plural noun phrase, which is a set of sets, and the denotation of the determiner, which ranges over atoms. To resolve this mismatch, Winter proposes the following type-shifter called *dfit*, which maps a determiner $D(P)(Q)$ to a relation $dfit(D)$ between sets of sets $\mathcal{P}$ and $\mathcal{Q}$:

$$(139) \quad dfit(D) =_{\text{DEF}} \lambda\mathcal{P}\langle\langle e, t \rangle, t \rangle. \lambda\mathcal{Q}\langle\langle e, t \rangle, t \rangle. D(\bigcup \mathcal{P})(\bigcup(\mathcal{P} \cap \mathcal{Q}))$$

The union operations ensure that the determiner relation is ultimately evaluated over the underlying atomic individuals rather than over the pluralities themselves. For example, *Exactly two judges met in the deliberation room* counts two individual judges and not two pluralities of judges. If we apply this operator to existential and universal singular quantifiers, we get the following denotations, where $dfit(\text{every})$ is synonymous with *all*:

$$(140) \quad \begin{array}{ll} \text{a.} & \llbracket \text{some} \rrbracket = \lambda P\langle e, t \rangle. \lambda Q\langle e, t \rangle. P \cap Q \neq \emptyset \\ \text{b.} & \llbracket dfit(\text{some}) \rrbracket = \lambda\mathcal{P}\langle\langle e, t \rangle, t \rangle. \lambda\mathcal{Q}\langle\langle e, t \rangle, t \rangle. (\bigcup \mathcal{P}) \cap (\bigcup(\mathcal{P} \cap \mathcal{Q})) \neq \emptyset \end{array}$$

$$(141) \quad \begin{array}{ll} \text{a.} & \llbracket \text{every} \rrbracket = \lambda P\langle e, t \rangle. \lambda Q\langle e, t \rangle. P \subseteq Q \\ \text{b.} & \llbracket dfit(\text{every}) \rrbracket = \lambda\mathcal{P}\langle\langle e, t \rangle, t \rangle. \lambda\mathcal{Q}\langle\langle e, t \rangle, t \rangle. (\bigcup \mathcal{P}) \subseteq (\bigcup(\mathcal{P} \cap \mathcal{Q})) \end{array}$$

Assume, contrary to Winter (2002a), that there is no lexical distinction between atom and set predicates, so that the flat set $\{a, b, c, d, e\}$ may itself be in the extension of *convict the defendant*. Further assume that no proper subset of this set can be in the extension of the predicate, since only the five judges together can convict the defendant. Under these assumptions, (138-a) states that some judge is a member of a plurality that convicted the defendant, while (138-b) states that every judge is a member of such a plurality. However, since no proper subset of $\{a, b, c, d, e\}$ can satisfy the predicate, the relevant plurality must be the full set in both cases. The two sentences are therefore contextually equivalent, leading to a violation of NTA.

More generally, in contexts where only the maximal plurality can satisfy the predicate, contextual equivalence arises under any analysis that construes plural quantification in terms of participation. For universal quantification, this means that instead of requiring every individual in the restrictor to satisfy the predicate individually, it suffices that every such individual be part of some plurality that satisfies the predicate. If the only such plurality is the maximal one, existential and universal quantification necessarily converge. The proposed account thus does not hinge on the particular semantics for *all* adopted here.

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